



TM292

MapleSim and Functional Mock-up Interface (FMI)

Requirements

Training modules:	TM210 - Working with Automation Studio TM213 - Automation Runtime TM223 - Automation Studio diagnostics
Software	B&R Automation Studio 4.6.x.x or higher Scene Viewer 3.1.8.0 or higher Maple 2019.2 or higher MapleSim 2019.2 or higher MapleSim - CAD toolbox MapleSim - B&R toolboxes ServoSoft 3.3.518 or higher

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1 Introduction

The word simulation comes from the Latin *simulatio* = pretense. In today's technology, however, simulation is understood as the reproduction of a real system by a computer program. The accuracy and extent with which the real system is reproduced depends heavily on the requirements and the available computing power. The more accurate the reproduction of the real system is, the better it reflects reality.

Simulation models are created for various reasons in a technical environment. [Fig. 1](#) shows some intended areas of use. One important reason is analyzing the systems. Creating a simulation will strengthen your understanding of the system, process or machine. The gained knowledge can be used to better control the system. From an economic point of view, a better understanding of the system leads to an increase in efficiency. Another reason for using a simulation is the ability to experiment. Destructive and non-destructive testing can both be carried out in the simulation without high costs. An important aspect here is, for example, checking the drive sizing. Drives can also be sized using the simulation model. One area that has become increasingly important in recent years is software testing. Machine and process software can be tested against a digital twin (simulation model). Early detection of software errors can significantly reduce commissioning times on site. The digital twin can also be used for predictive maintenance. The simulation runs parallel with the real system in real time; if the deviation is too large, this indicates a malfunction. Using a digital twin, such malfunctions can be detected at an early stage. Further reasons for creating a simulation model include: Presentations, operator training, usability tests, ergonomic questions, etc. .

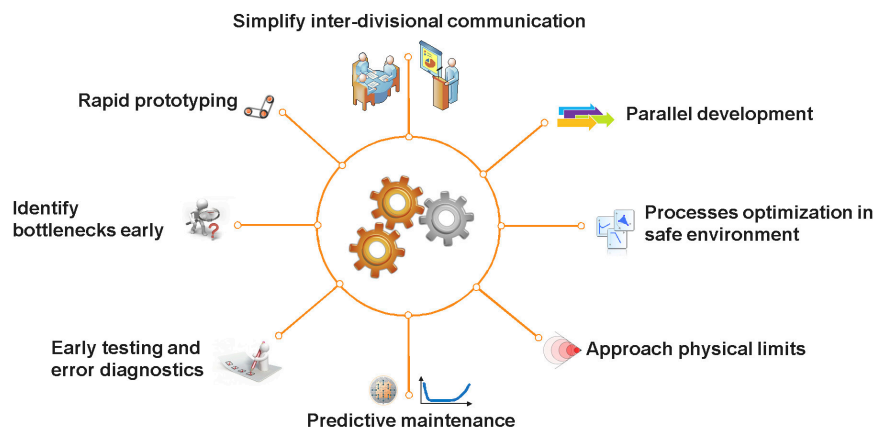


Figure 1: Use of simulation models

Further information and explanations related to the basics and virtualization of control technology can be found in "**TM291 - Basics of virtualization and simulation for industrial control technology**".



TM291 - Basics of simulation for industrial control technology

1.1 Learning objectives

This training module uses selected examples illustrating typical application tasks to help participants learn how various functions in B&R Automation Studio are represented and configured.

- Participants will learn which simulation tasks MapleSim is best suited for.
- Participants will learn how to use the MapleSim program.
- Participants will become familiar with the following MapleSim libraries: **Multibody**, **1D Mechanical** and **Signal Blocks**.
- Participants will become familiar with the following MapleSim apps: **B&R Automation Studio FMU Generation**, **B&R SERVOfsoft Data Generation** and **1-D Motion Generation**.
- Participants will learn how to create models of mechanical systems without deriving differential equations.
- Participants will learn how to export a MapleSim model as a Functional Mock-Up (FMU) and then import it into B&R Automation Studio.
- Participants will learn how to use the B&R Scene Viewer to visualize the MapleSim model in B&R Automation Studio.

An advantage of pre-simulation of physical systems is the possibility to size the drives. If the system is known and is simulated with the maximum degree of dynamics, the calculated forces and torques can be used to determine the minimum sizes of the drives. The **B&R SERVOfsoft Data Generation** app in MapleSim in combination with the SERVOfsoft software is a great help here.

2 Simulation

The typical B&R simulation levels, which differ in their level of detail, are shown in Fig. 2. A distinction is made between the automation hardware level, the component and machine level and the process and plant level. "TM291 - Basics of simulation for industrial control technology" provides a good overview of the different levels.



TM291 - Basics of simulation for industrial control technology

MapleSim is best suited for machine and component simulation. With MapleSim, physical quantities such as positions, velocities and forces can be very accurately calculated. For this reason, MapleSim is also well suited for dimensioning. In general, hardware components can also be modeled with MapleSim, but it makes sense to use the already existing simulation models in B&R Automation Studio.

Other tools are also better suited for plant and process simulation. One reason is the handling, another the required processing power.

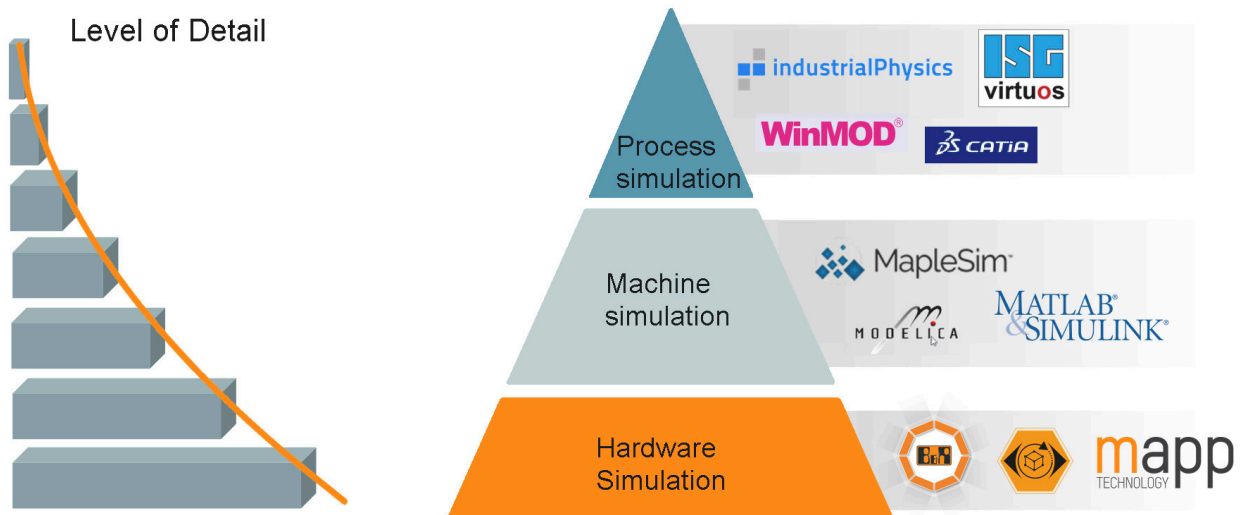


Figure 2: Simulation levels

2.1 Preparation

Choosing the right simulation tool is by no means easy. At the beginning, you should define where you want to use the simulation. A few areas of application are shown in Fig. 1.

MapleSim is well suited for the following applications:

- Process optimization in a safe environment: Sizing drives, developing control algorithms
- Rapid prototyping: Quickly creating different system variants e.g. different mechanical geometries
- Determination of system load limits: Determining the maximum overload time
- Predictive maintenance: Operating a digital twin that runs parallel to a real system -> early error/wear detection.
- Simplification of interdisciplinary communication: Using visualization, it quickly becomes clear whether the system was modeled as originally intended.

3 MapleSim basics

Compared to other simulation tools, using MapleSim together with Modelica is a very efficient modeling method (see Fig. 3) . Unlike the conventional method, a MapleSim model does not require differential equations. This saves time and reduces the probability of errors. There is more time to build different system variants and test them against each other, for example.

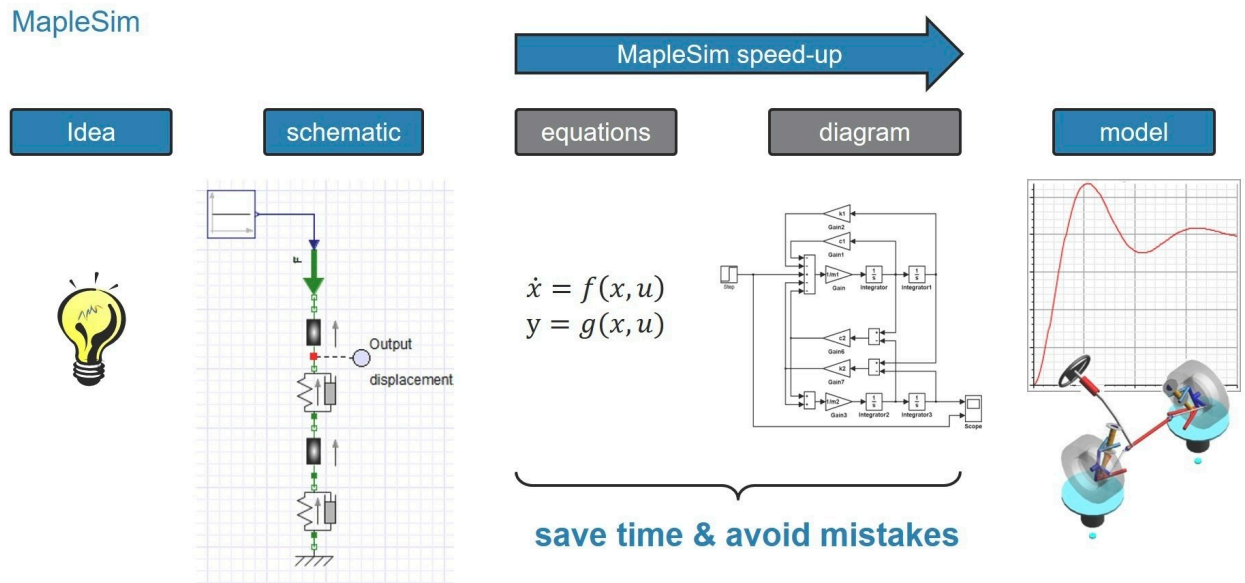


Figure 3: Maple workflow

Systems theory and modeling

Initial value problems

The solution of an initial value problem is the solution of the differential equation with additional consideration of a specified initial value.



https://en.wikipedia.org/wiki/Initial_value_problem

In mathematics, physical models can very often be mapped with the help of standard differential equation systems. An example in mechanics is a spring-mass oscillator or in electrical engineering an RLC circuit. Usually, these differential equation systems contain derivatives of second order state variables (which can be converted into a first order state equation system). This system can become very complex, however, which means that its solution is no longer easy to calculate analytically. Numerical solution algorithms are then used (one-step or multi-step methods), of which the best-known examples are the Euler method or the Runge-Kutta method.

FMI (Functional Mock-up Interface) / FMU (Functional Mock-up Unit)

The **Functional Mock-up Interface** (FMI) defines a standardized interface that can be used to couple different simulation software. A physical system (e.g. an industrial plant) can consist of several components that can come from different manufacturers. This is where the FMI comes into play. The idea is to enable a variety of components to interact with each other in a complex way by merging and displaying them in a simulation.

Experience has shown that the implementation of FMI via a software modeling tool makes it possible to create simulation models that can be coupled with each other.

An example model based on a vehicle could look like this: Software A handles the engine, Software B handles the gear-box, Software C handles the controls, Software D handles ... With FMI, these various software models can easily be coupled and combined to a complete model.

Another possibility is to create a software library called FMU (**F**unctional **M**ock-up **U**nit). The FMI specifications are provided as open source licenses. Each FMU model is made available as a ZIP file with extension ".fmu" and includes the following components:

- an XML file describing, among other things, the definition of the variables used by the FMU;
- all formulas used in a model (defined as C functions);
- optional additional data such as parameter tables, the user interface, documentation required by the model, etc.



https://en.wikipedia.org/wiki/Functional_Mock-up_Interface#Architecture

Modelica

Modelica is an object-oriented modeling language for physical models. There are different graphical development environments for this language (e.g. MapleSim or Dymola). They allow the user to develop complex simulation models using graphical symbols, each representing a physical object. Objects are connected using connectors, which are also usually non-directional, i.e. bidirectional. An example of this is a motor shaft. With the motor shaft, the motor can set a gearbox in motion. The load of the gearbox also affects the motor via the motor shaft, however, and the motor shaft thus has 2 directions of action, one from the motor to the gearbox and vice versa.

Modelica is suitable for describing interdisciplinary problems in a wide range of fields such as mechanics, electrical engineering and electronics, thermodynamics, hydraulics and pneumatics, closed-loop control and process control. The language definition and the Modelica standard library are freely available and are further developed and promoted by the Modelica Association.

A major advantage of Modelica is that it works with equations instead of assignments. Variables being searched for do not need to be resolved. Another advantage is that variables can have properties assigned (physical size, unit). This allows verification of equations by the simulation software.



<https://en.wikipedia.org/wiki/Modelica>

Computing time

The **cycle time** of the numerical solver in MapleSim must match the **task class cycle time** of B&R Automation Studio to achieve correct simulation. Setting a lower time results in a more accurate calculation and simulation of the physical model. But this also means that computing time can increase considerably. It is necessary here to choose a time that reproduces the simulation with sufficient accuracy while keeping the computing time as low as possible.

3.1 User interface

The following figure provides an overview of the basic structure of MapleSim.

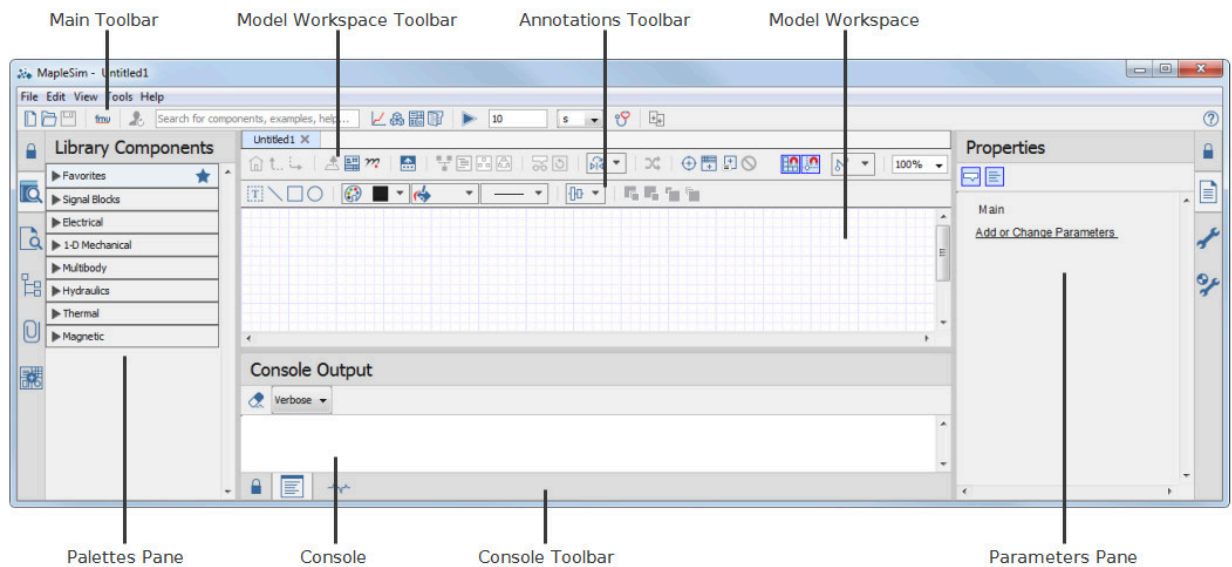


Figure 4: MapleSim user interface

Component	Description
Main toolbar	Includes the tools to run a simulation, visualize its results, search MapleSim for help and perform other simple tasks.
Model workspace toolbar	Includes tools to hierarchically browse the model and its subsystems, change the model view, view the associated MODELICA code, group components and add measurement sensors.
Annotations toolbar	Includes tools to add annotations and arrange objects.
Model workspace	Allows the model to be built and edited in block diagram view.
Palettes pane	Includes expandable menus that can be used to build the model and manage the MapleSim project. 5 tabs can be used: - Library components: Ready-made models and domain-specific components can be added to the model. - Local components: Entries from the subsystem and self-made components can be inserted. - Model tree: Here you can navigate through the model with the help of a structure tree. - Attached files: Attachments for the model can be seen here: Documents, parameter sets and CAD drawings. - Add apps or templates: Contains entries with ready-made tools to build or analyze the model.
Console	Buttons on the Console toolbar can be used for visualization to display the following: - Console output: Displays all associated messages that reflect the status of the MapleSim engine during a simulation. There is also the possibility to delete the messages. - Diagnostics information: Shows all diagnostic messages for debugging while building the model. The subsystem where an error occurs can be identified.
Console toolbar	The message type can be selected.

Component	Description
Parameters pane	Contains the following entries: - Properties: Properties of model components such as names, property values, initial conditions and measurement sensor properties can be viewed and modified. - Simulation settings: Simulation options such as the duration of the simulation or optional parameters for the solver and the simulation engine. - Multibody settings: Here you can specify options for visualizing the multibody components. The content of this tab changes depending on the choice in the Model workspace.

The simulation is started when clicking on **Play** . If the simulation was successful, the **Show simulation results** icon  flashes. By clicking on this symbol, the resulting visualization of the simulation can be viewed. The configured plots are also available in this window.

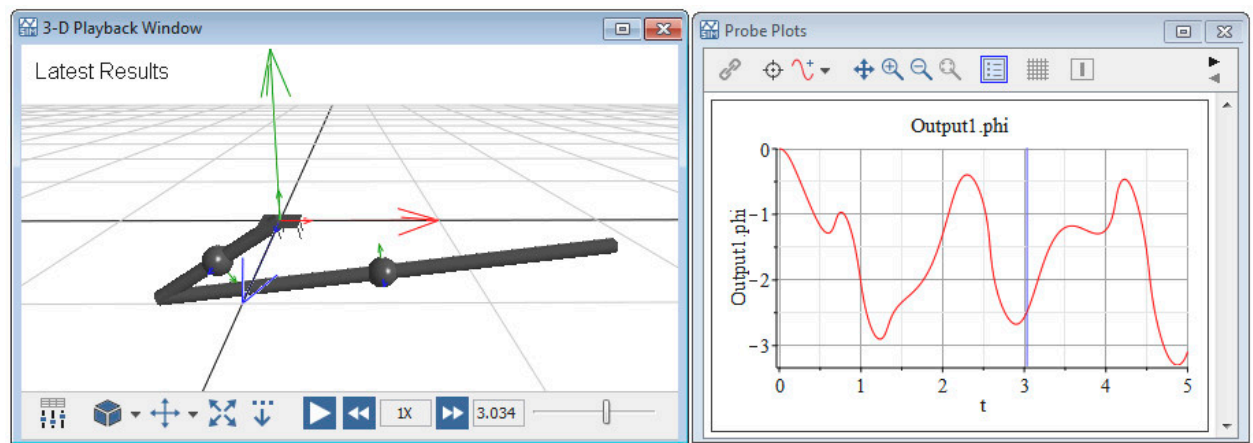


Figure 5: Simulation results and plots for a double pendulum model as an illustrated example.

3.2 Help documentation

All functions offered by MapleSim are described in the **MapleSim help documentation**. To get help for a specific element, press the F2 key.
Help documentation is also available online at www.maplesoft.com.
At the beginning, it is also worthwhile to view youtube video **Getting Started with MapleSim** .



Quick help related to various objects in MapleSim can be called up by pressing the F2 key.

4 Modeling

To see how MapleSim helps to simplify physical models and to see how the necessary mathematics disappears into the background, a simple model of a mathematical pendulum will be created. First, the mathematical model is derived by hand and then MapleSim is used.

It becomes clear how much MapleSim takes over the steps necessary to perform a complete simulation. For more complex systems, the effort required to manually derive the equations quickly becomes very large. This can still be kept to a minimum using the Modelica approach.

4.1 Mathematical pendulum

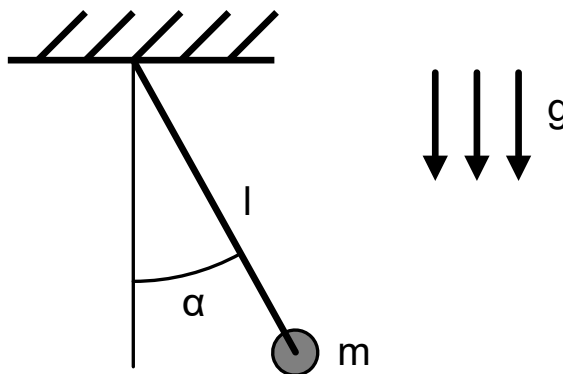


Figure 6: Mathematical pendulum, sketch

Deriving the motion equation with Newton

The following derivation is made using Newton's 2nd law:

$$F = m \cdot a$$

Mass m can only move along the arc (t = tangential). F describes the force and a the acceleration. The motion equation for the mathematical pendulum is written as follows:

$$F_t = m \cdot a_t$$

It is important that tangential acceleration a_t acts in the same direction as force F_t (see [Fig. 7](#) and [Fig. 8](#)). If the directions are different, this must be taken into account by the sign.

[Fig. 7](#) shows that a_t can be expressed using angle α . α is measured in the opposite direction as a_t , thus the negative sign.

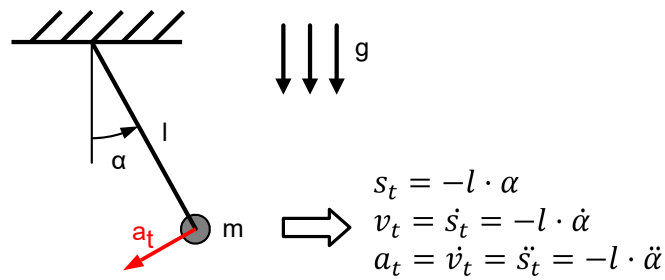


Figure 7: Mathematical pendulum: Position, velocity and acceleration

The tangential force F_t can also be described using weight $F_G = m \cdot g$ and angle α .

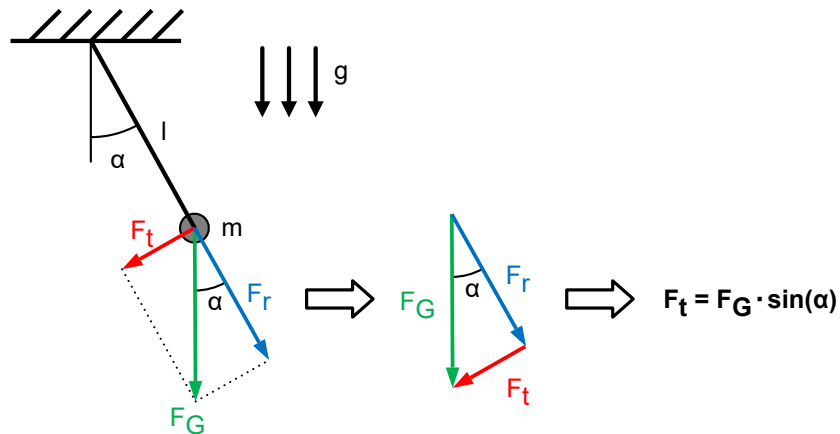


Figure 8: Mathematical pendulum: Forces

When used in equation $F_t = m \cdot a_t$, this results in the following (with l as pendulum length and g as acceleration due to gravity):

$$m \cdot g \cdot \sin(\alpha) = m \cdot -l \cdot \ddot{\alpha}$$

Converting the equation results in the differential equation for the mathematical pendulum:

$$\ddot{\alpha} + \frac{g}{l} \cdot \sin(\alpha) = 0$$

Deriving the motion equation with Lagrange

Lagrange mechanics can also be used for derivation as an alternative to Newton. In Lagrange mechanics, the generalized coordinates must first be determined. This is a minimal set of independent coordinates to unambiguously describe the spatial state of the system.



https://en.wikipedia.org/wiki/Generalized_coordinates

Lagrange mechanics is explained in the form of an example in the following. First, the Lagrange function L - with the help of the kinetic energy E_{kin} and the potential energy E_{pot} - is determined as follows:

$$L = E_{kin} - E_{pot}$$

Then this function is derived according to the generalized coordinates q and time (Lagrange equation):

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i} = 0$$

For the mathematical pendulum, α is defined as a generalized coordinate. This choice is generally not easy and there are systems where it is not possible to make such a choice. In any case, the system must be holonomic.



https://en.wikipedia.org/wiki/Lagrangian_mechanics

The Lagrange function is then set up:

$$L = E_{kin} - E_{pot} = \frac{1}{2} m (l \cdot \dot{\alpha})^2 - m \cdot g \cdot l \cdot (1 - \cos(\alpha))$$

The function is derived to obtain the Lagrange equation:

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\alpha}} = \frac{d}{dt} (m \cdot l^2 \cdot \dot{\alpha}) = m \cdot l^2 \cdot \ddot{\alpha}$$

$$\frac{\partial L}{\partial \alpha} = -m \cdot g \cdot l \cdot \sin(\alpha)$$

$$m \cdot l^2 \cdot \ddot{\alpha} + m \cdot g \cdot l \cdot \sin(\alpha) = 0$$

Simplification of Lagrange equation results in the motion equation (identical with Newton):

$$\ddot{\alpha} + \frac{g}{l} \cdot \sin(\alpha) = 0$$

4.2 Modeling in MapleSim

4.2.1 Multibody library

The **Multibody library** is introduced in the following section. This library provides blocks for modeling 3-D mechanics. Bodies, masses and joints, among other things, are made available.

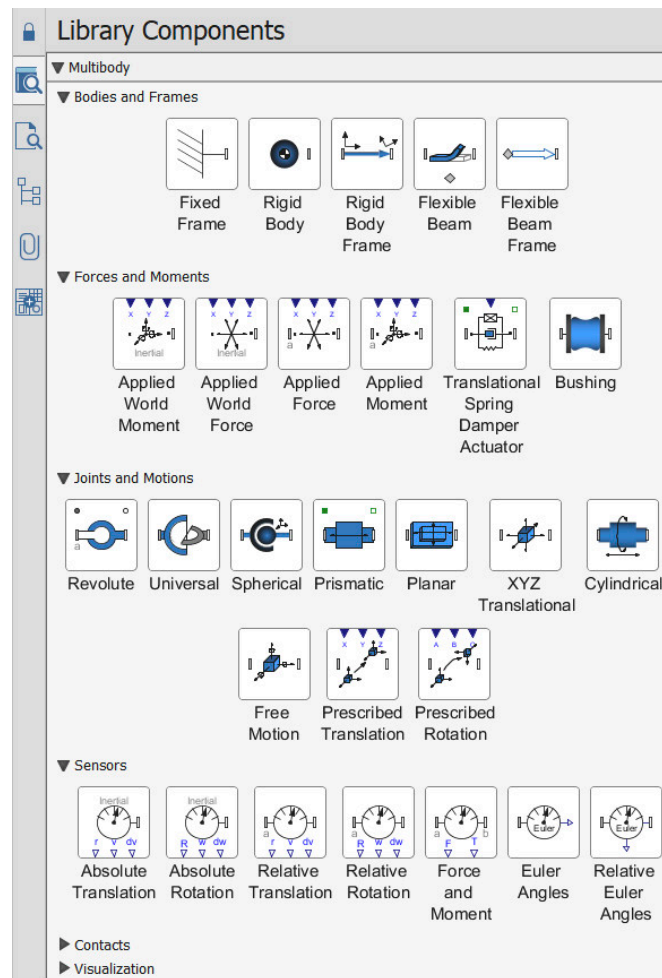


Figure 9: Multibody library

4.2.2 Mathematical pendulum

Creating the model with MapleSim

In MapleSim, the following elements from the **Multibody library** will be used:

A **fixed frame** to fix the pendulum in space:



A **revolute** (hinge joint), to allow rotational freedom:



A **rigid body frame** instead of the rope:



A simplification is already being made here. The rope of the pendulum is assumed to be always tensioned. The rope can therefore be replaced by a rigid body. In manual derivation, this simplification is implicitly made by the choice of the coordinate system.

A **rigid body**: This represents a mass within a system. In the properties for this object, you can also configure a rotational inertia.



The individual elements can be dragged and dropped in the **workspace**. These can then be connected to each other using flanges (i.e. connection points). The finished model looks like this in the **workspace**:

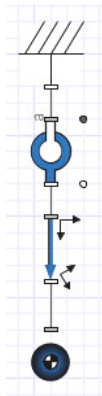


Figure 10: Mathematical pendulum in MapleSim



With the help of the shortcut menu (right mouse button), elements can be rotated and flipped in the **workspace**.

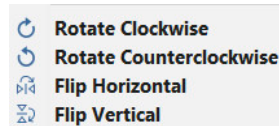


Figure 11: Rotating and flipping elements

Now the simulation can be started  and the result reviewed .
The model is displayed in the **3D playback** window as follows:

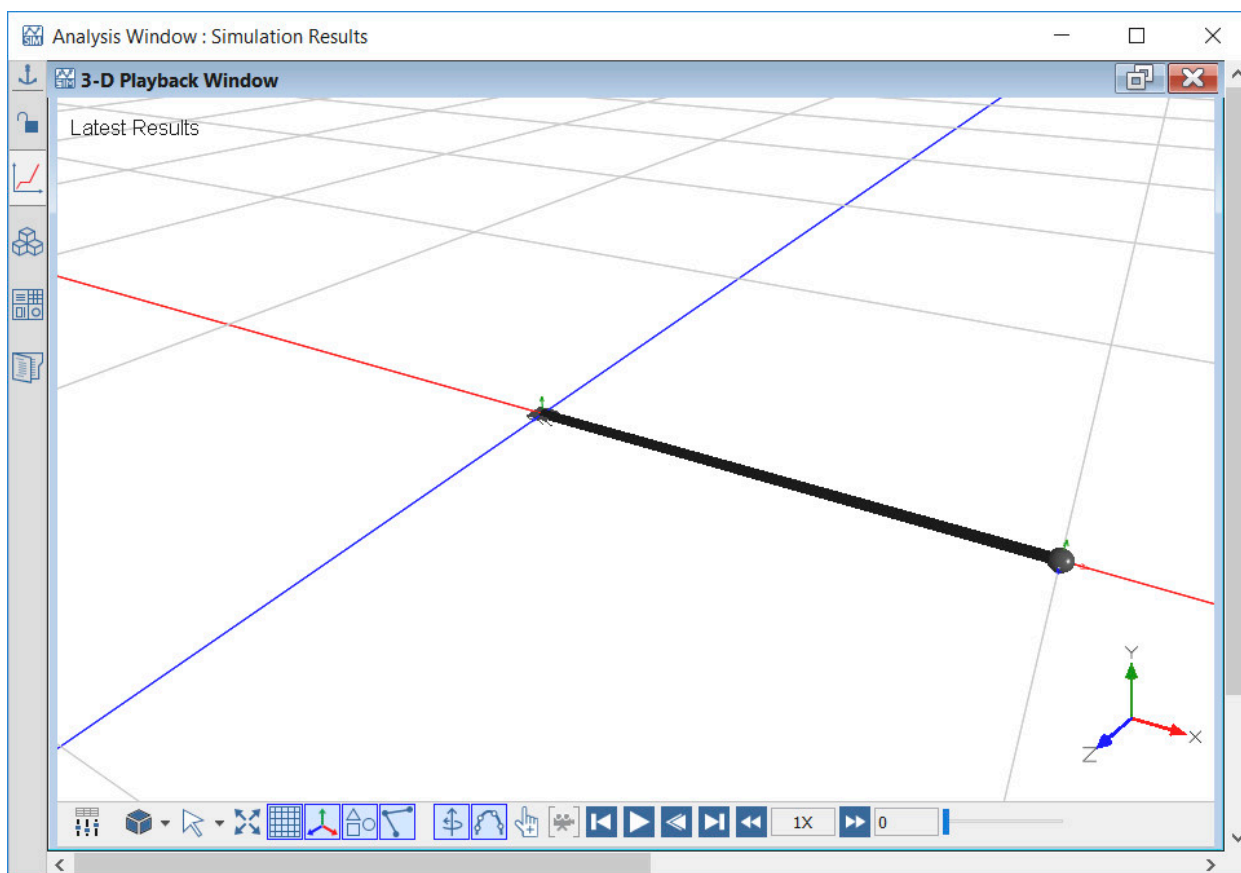


Figure 12: MapleSim 3D playback window

It can be seen that the pendulum starts with a deflection of 90° . This is based on default values of the inserted library elements. In addition, the pendulum swings in the x-y plane when the coordinate system is viewed from the bottom left.



The F2 key can be used to open the help documentation for each element.

What is necessary to make the pendulum swing in the y-z plane should be indicated in the form of an example. First, the properties of the hinge joint are observed:

Parameters	
$\hat{e}_1$	[0,0,1]
K_s	0 $\frac{N \cdot m}{rad}$
ϕ_0	0 rad
K_d	0 $\frac{N \cdot m \cdot s}{rad}$
$IC_{e,\omega}$	Ignore
θ_0	0 rad
ω_0	0 $\frac{rad}{s}$

Figure 13: Hinge joint parameters

$\hat{e}_1$ indicates the rotation axis of the joint. In this case [0,0,1] on the z axis. If the pendulum should swing in the y-z plane, the joint must be able to rotate around the x-axis. Therefore $\hat{e}_1=[1,0,0]$ is set. The other parameters will not be observed for the time being because they are not required here.

In addition, the parameters for the rigid body must be observed:

Parameters		
$\vec{r}$	[1, 0, 0]	m
Rotation Method	Euler Angles	
Euler Sequence	[1, 2, 3]	
$\vec{\theta}$	[0, 0, 0]	rad

Figure 14: Rigid body parameters

$\vec{r}$ indicates the direction in which the rigid body is extended. [1,0,0] m means the rod is extended 1 m in the x-direction. This is changed to $\vec{r} = [0,0,1]$. When the simulation is started again, the pendulum now swings in the y-z plane.



In general, all positions and expansions of an element are described in the coordinate system for flange a.. A detailed explanation can be found in the MapleSim help documentation (F2).

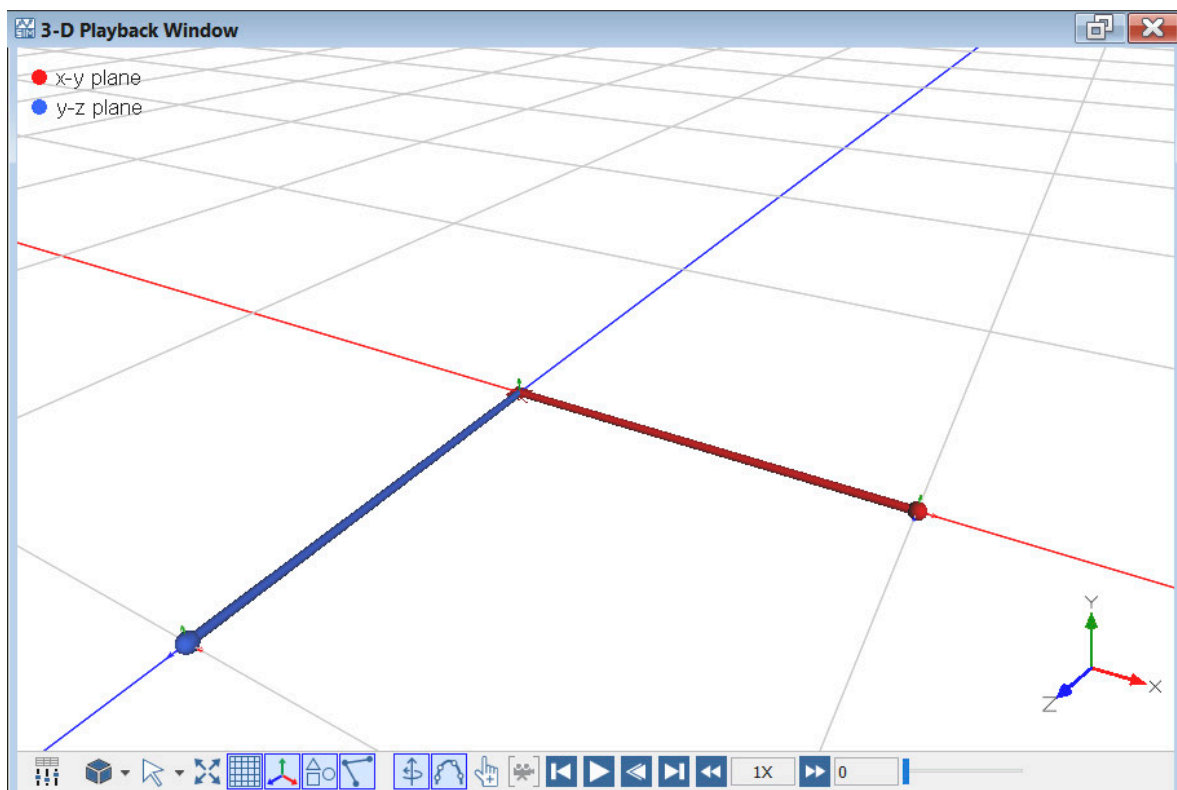


Figure 15: Mathematical pendulum simulation

You can also use **Equation extraction** (under **Add apps or templates**) to output the system's motion equations .

$$\left\{ \frac{l m \left(100 \left(\frac{d^2}{dt^2} \alpha(t) \right) l + 981 \cos(\alpha(t)) \right)}{100} = 0 \right\}$$

Figure 16: Automatically generated motion equation

Further simplification results an identical motion equation, which was already derived manually above.

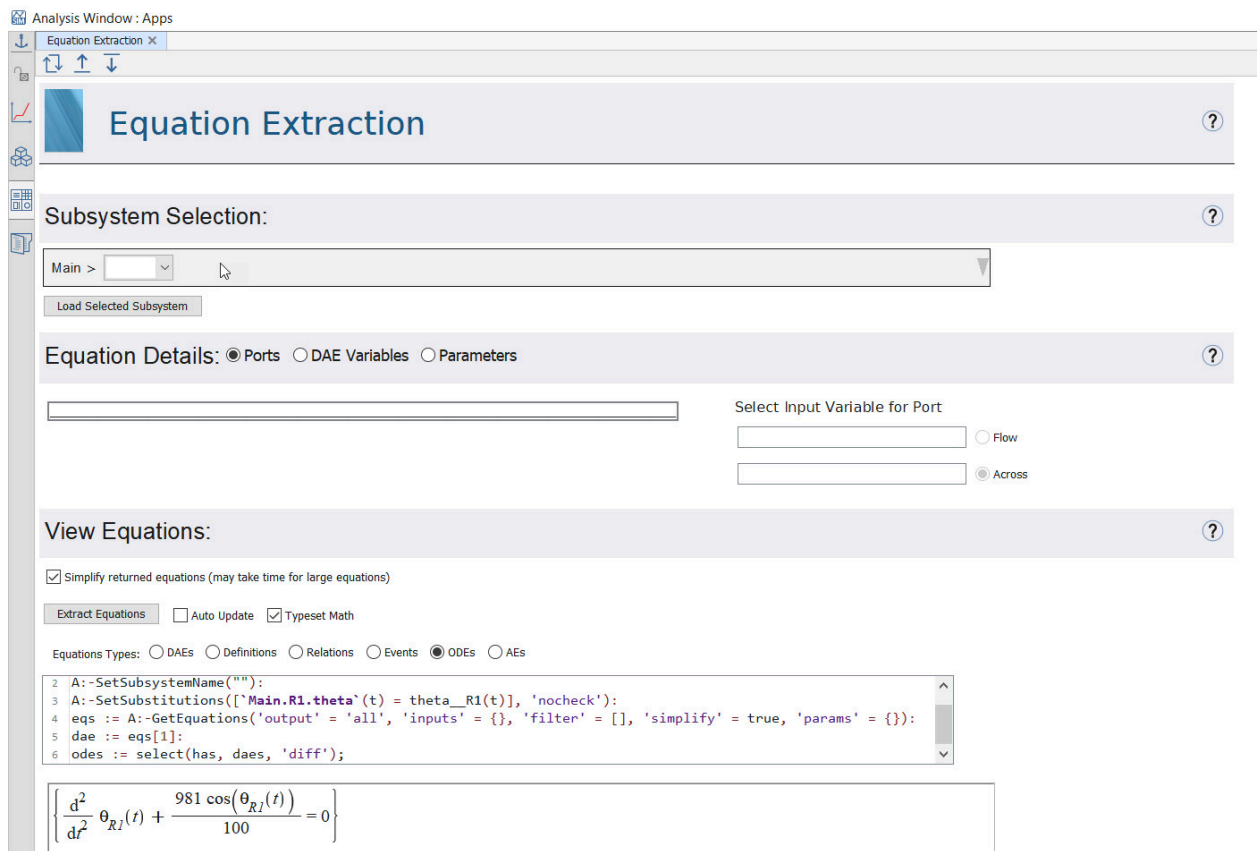


Figure 17: Equation extraction from MapleSim.

Modeling a mathematical pendulum

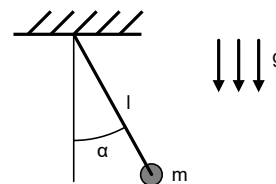
Create a simulation model of a mathematical pendulum

Use the following values:

$m = 1 \text{ kg}$

$l = 1 \text{ m}$

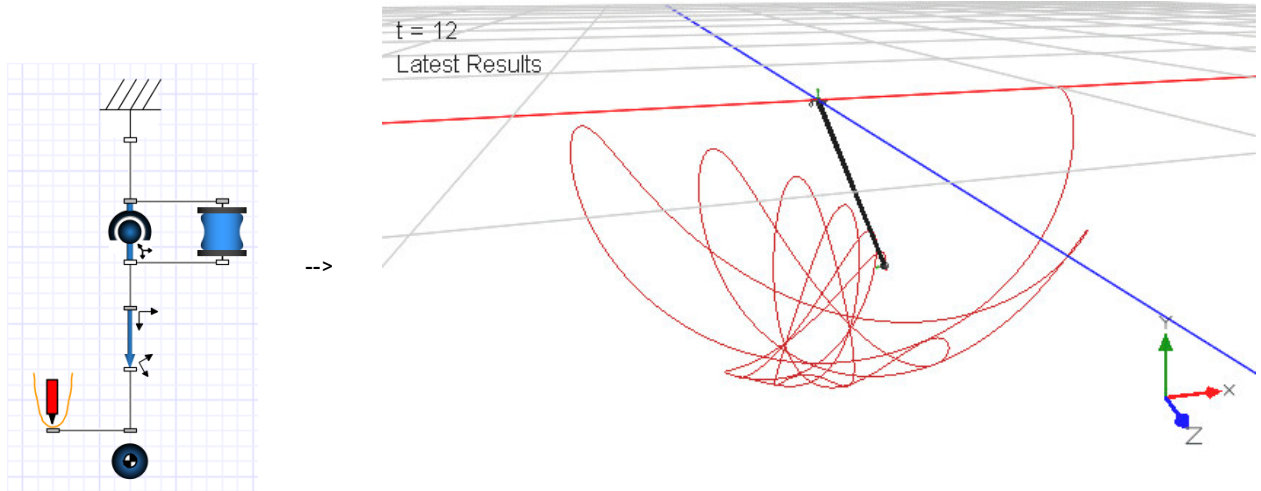
$g = 9.81 \text{ m/s}^2$



- 1) Familiarize yourself with the **Multibody library**.
- 2) Build the model and start the simulation.
- 3) Convert your model so that the pendulum swings in a different direction. As depicted in [Fig. 15](#).
- 4) Adapt your model so that the pendulum can swing in all three spatial directions.
Tip: Use a spherical joint.
- 5) Set the initial conditions for the point mass so that the pendulum performs true 3D movement.
Tip: Change the initial conditions in the parameters for the mass. Make sure that $\vec{r}_0$ matches the dimensions of the rod and introduce an initial speed to the system using $\vec{v}_0$. Set the initial conditions IC_{r,v} to **Strictly enforce**.

- 6) Record the path of the mass. Tip: Use the **Path Trace** element in the **visualization** area of the **Multibody library**.
- 7) Dampen the oscillation of the pendulum. Tip: Use a bushing.

Modeling results



5 Creating the initial Functional Mock-up Unit (FMU)

To be able to use a model from MapleSim in B&R Automation Studio, a Functional Mock-up Unit (FMU) must be created from it. An FMU should be created from the pendulum model. The following steps are necessary for this:

- 1) Create a subsystem from an existing model.
- 2) Install and define the mechanical drives.
- 3) Specify a solver (e.g. **fixed step solver**, sampling rate 400 μ s).
- 4) Export the model as an FMU with **RK4 solver**.

Tools used

- B&R Automation Studio FMU generation

5.1 The pendulum model

The existing pendulum model (see Fig. 10) should be converted into an FMU.

5.2 Subsystems

As the first step, a subsystem must be formed from the model (see Fig. 18), which is later converted into an FMU. Right-click with the mouse to select and convert all necessary elements into a subsystem.

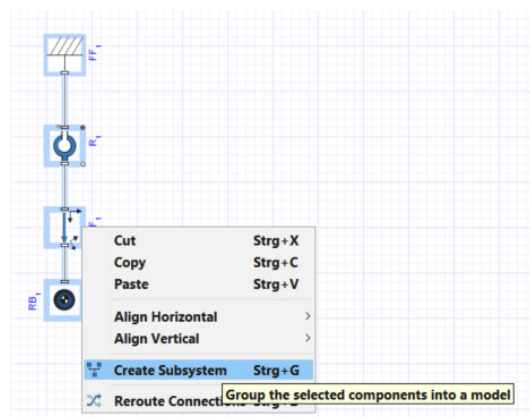


Figure 18: Pendulum model as a subsystem

In the subsystem, inputs and outputs must then be defined that will later serve as an interface in B&R Automation Studio. It should be noted here that these must be unidirectional, which is indicated by the arrows (see Fig. 19). To achieve this, additional elements from the 1D mechanical library are required. On the input side, element **Torque** from the **Torque Drivers** library and on the output side, **Angle Sensor** from the **Sensor** library.

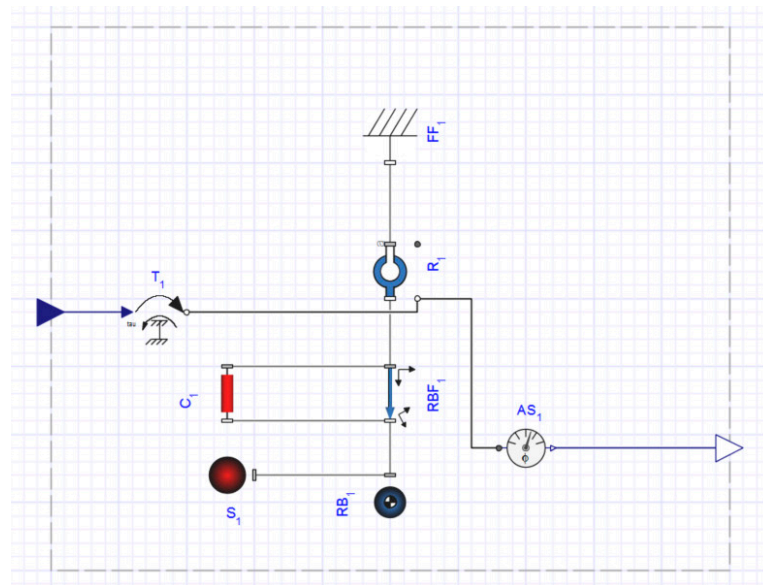


Figure 19: Pendulum subsystem with inputs and outputs

This subsystem can then be integrated and tested in a MapleSim simulation (see Fig. 20).

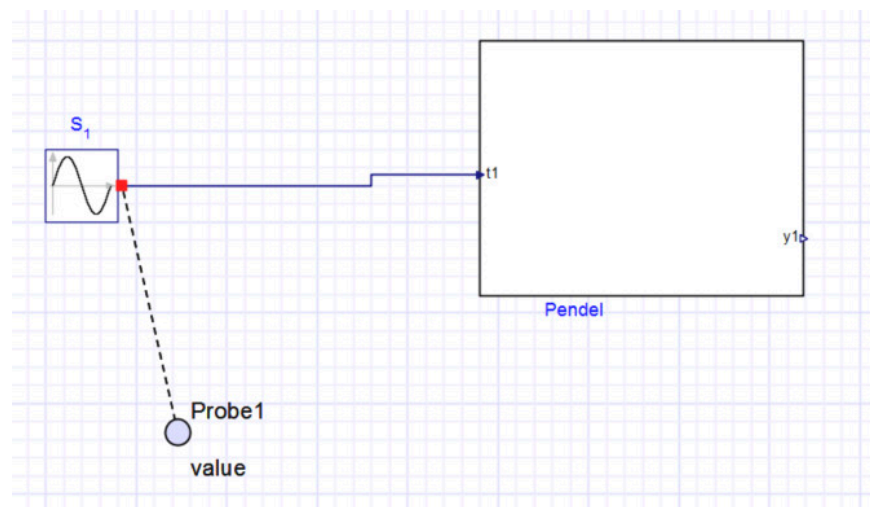


Figure 20: Subsystem pendulum integrated in a MapleSim simulation

Building a subsystem

Convert the existing pendulum model into a subsystem.

5.3 Tracing signals

With the aid of measurement sensors (**probes**), signals from the model can be visualized allowing the model itself to be verified. The measurement sensors are shown in the **Model Workspace toolbar**. The pendulum results are shown in [Fig. 21](#) in the form of an example.

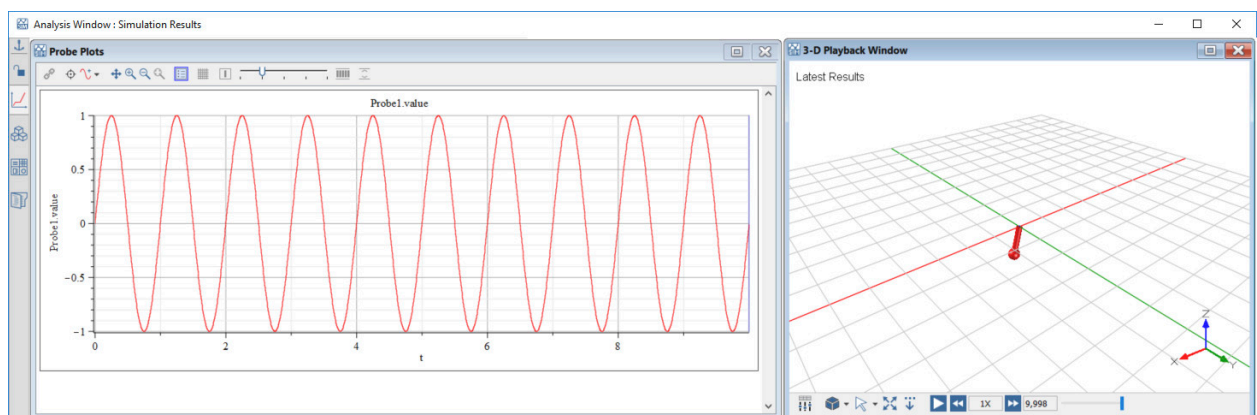


Figure 21: The results of the simple pendulum with measurement sensor.

Tracing signals

Experiment with the model and its physical quantities. Output various measured values, such as position, velocity or forces. Try to interpret these values and explain how you can tell that the system is modeled correctly.

5.4 Exporting the Functional Mock-up Unit from the pendulum model

Open **FMU export**, see Fig. 22. Then the model must be exported, see Fig. 23.

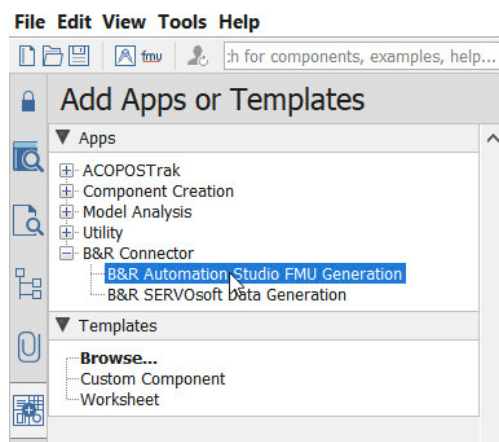


Figure 22: Open B&R Automation Studio FMU generation.



It is necessary to ensure that the step size is equal to the cycle time of the PLC. (see Fig. 23). If the step size deviates from the cycle time, correct behavior of the model in the simulation is no longer guaranteed.



Names for archives and models with more than 7 characters should be avoided.

If the program name has less than 10 characters, a connection to B&R Scene Viewer is established automatically.

If the function block has less than 20 characters, the connection to B&R Scene Viewer is established automatically.

When using more characters, the names in B&R Scene Viewer must be adjusted manually.

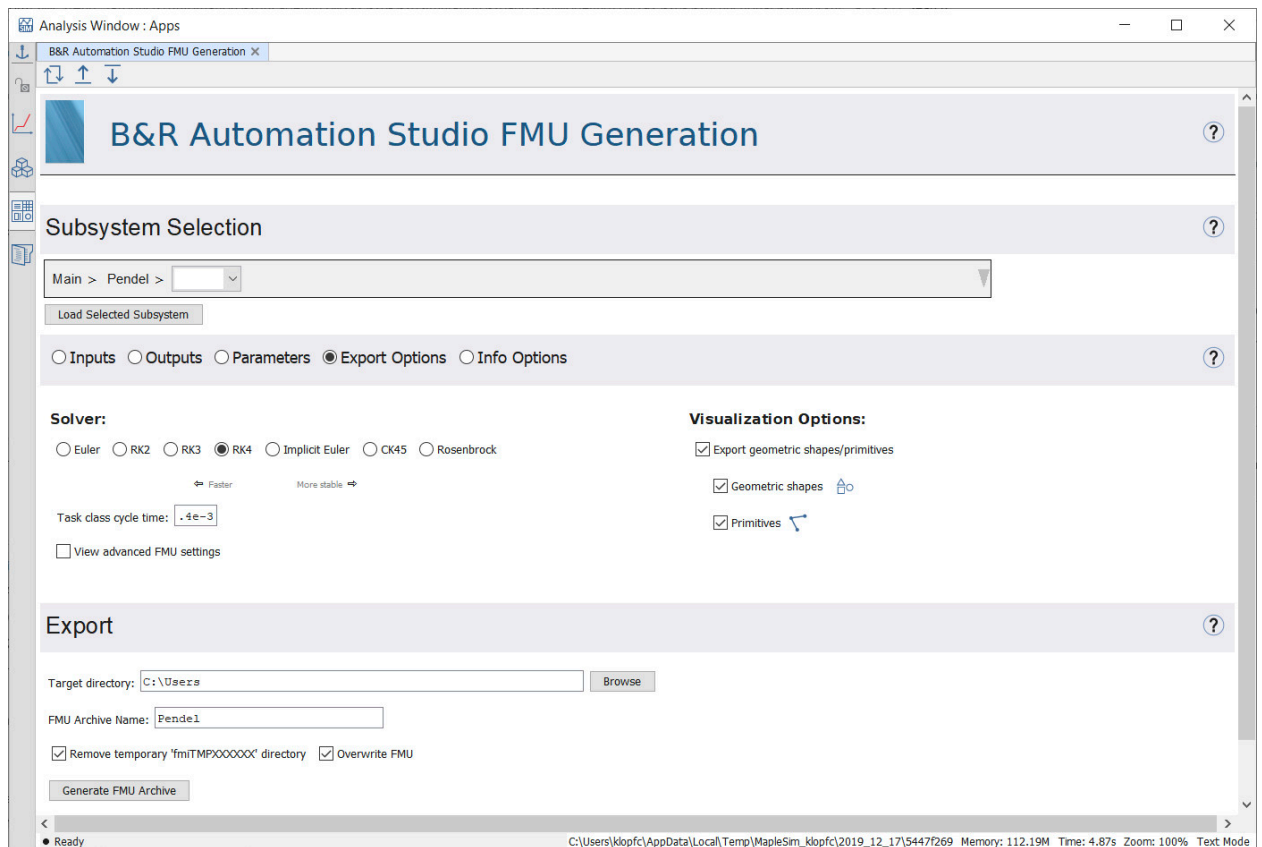


Figure 23: Parameters for FMU export of the pendulum.

After loading the subsystem, the following parameters in the **Export options** are important for creating an FMU (see Fig. 23):

- **Solver:** RK4 is recommended. The task class cycle time in which the task is called on the PLC must be entered in the field "Task class cycle time".
- **Visualization options:** The geometries of the model should also be exported.



The solvers: CK45 and Rosenbrock are variable step solvers. If this type of solver is used, the function block can determine its internal cycle time itself independent of the task class cycle time on the controller. Advantage: The model is more stable. Very short cycle times (less than 100 μ s) are also possible. Disadvantage: Unforeseen cycle time violations can occur because the computing time is variable. A cycle time violation leads to incorrect results.

5.5 Importing the FMU into a B&R Automation Studio project

First, the FMU library has to be loaded from the toolbox catalog (see Fig. 24).

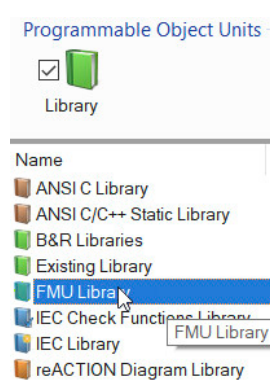


Figure 24: Adding an FMU in B&R Automation Studio.

The following wizard (see [Fig. 25](#)) can now be used to search for the exported FMU file. The remaining fields will then be filled in automatically.

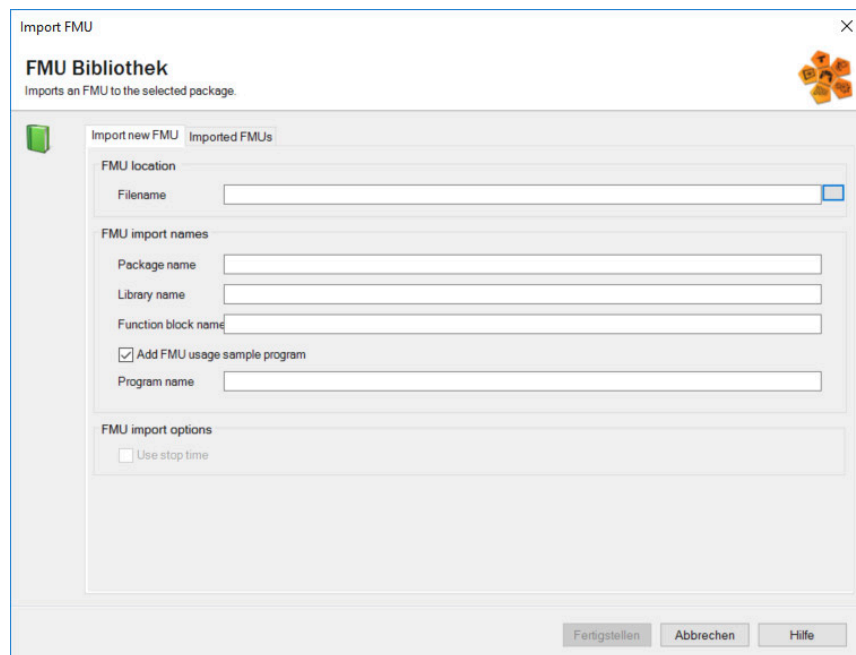


Figure 25: FMU library import wizard.

When the FMU has been imported into B&R Automation Studio, the task must be assigned to the corresponding **Cyclic#** (see right side [Fig. 26](#)). In addition, the time of the **Cyclic#** must be set according to the "Task class cycle time" that is set when exporting the FMU (in this case, 400 μ s). It may be necessary to adjust the System Tick.

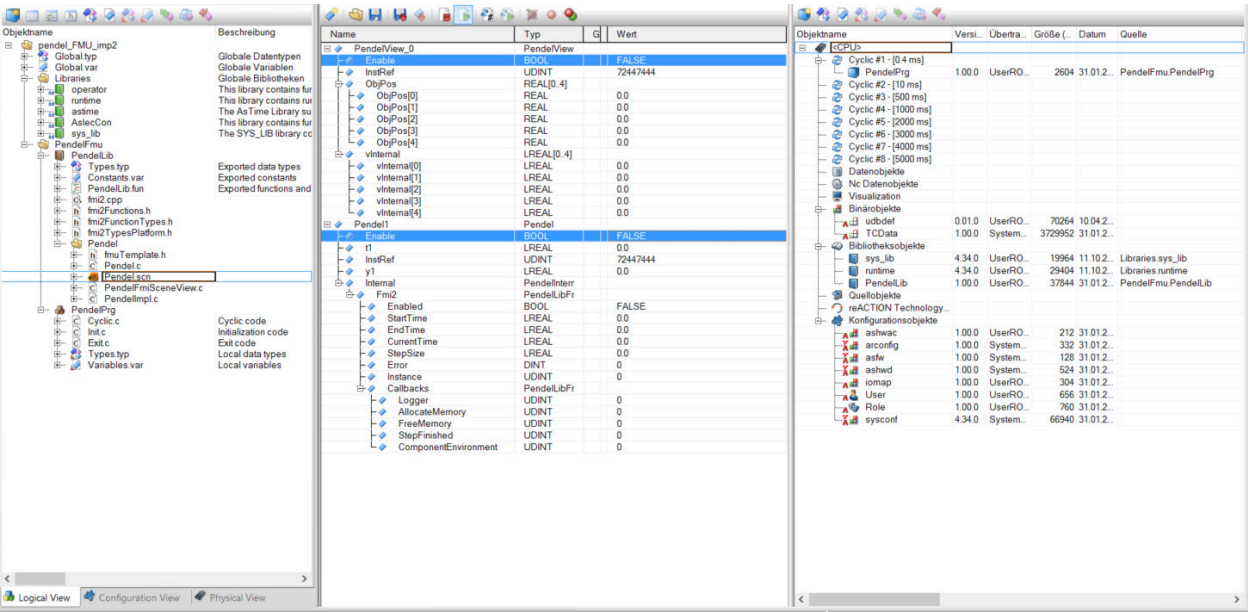


Figure 26: B&R Automation Studio GUI.

It is now possible to open B&R Scene Viewer (see Fig. 27) by double-clicking on **Pendel.scn** (left side Fig. 26).

5.6 B&R Scene Viewer

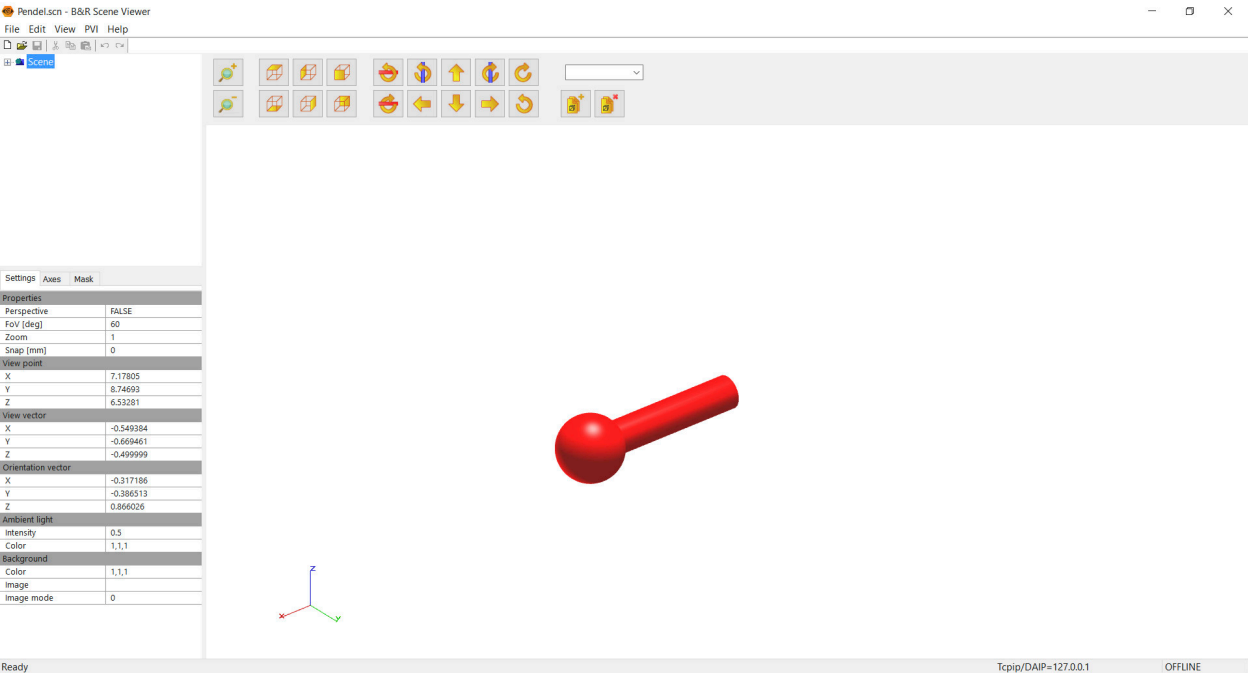


Figure 27: B&R Scene Viewer for the simple pendulum.

Via the PVI menu (see Fig. 28), it is possible to check if the correct settings have been made for a connection to B&R Automation Studio (see Fig. 29) and if the PVI connector is then started (see Fig. 28).

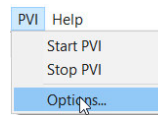


Figure 28: PVI tab in B&R Scene Viewer.

Figure 29: The settings for PVI.

As a final step, the simulation can be started via the Watch window in B&R Automation Studio by setting the Enable values of the visualization function block and the FMU function block to TRUE (see middle [Fig. 26](#)).

A pendulum movement is visible in B&R Scene Viewer.



In B&R Scene Viewer, you can use:

- The left mouse button to select an object,
- The middle mouse button + Ctrl to drag the camera and
- The middle mouse button + Shift + Ctrl to rotate the camera (around the selected object or virtual point).

6 CAD import into MapleSim

Importing STEP files

To import CAD files, the CAD toolbox must be opened. To do this, select menu option **File -> Import CAD...** (see [Fig. 30](#)).

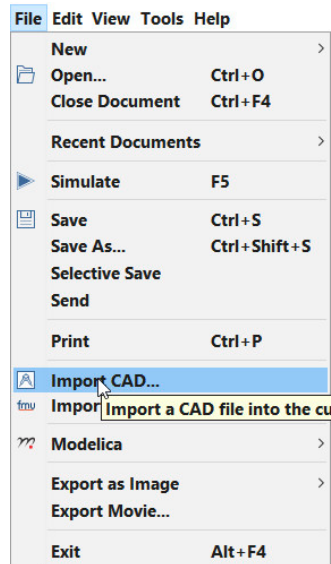


Figure 30: **Import CAD** in the **File** menu.

This opens the CAD toolbox (see [Fig. 31](#)).

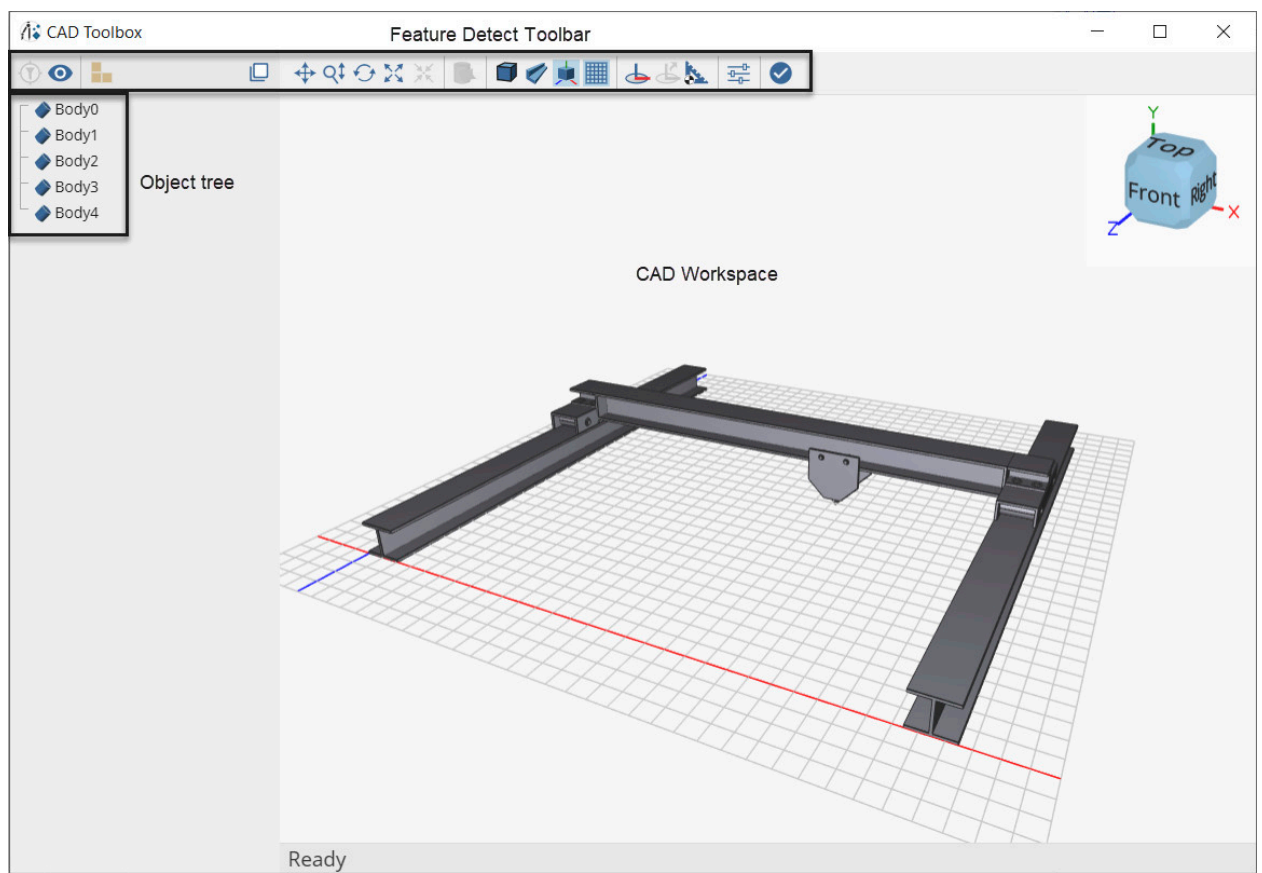


Figure 31: The **CAD toolbox**.

Important elements of the toolbox are shown in [Fig. 32](#).

The Import CAD Toolbar

In the Import CAD window, the following icons are found in the **Toolbar**:



Button	Description
	Toggle to coordinate tree view.
	Toggle to show only visible parts in tree view.
	Create a group.
	View properties.
	Pan the view.
	Zoom into or out from the CAD Workspace.
	Rotate the assembly.
	View the entire model.
	Focus view on the selected object.
	Show Geometry: All
	Show Geometry: Invalid
	Show Geometry: Only solid bodies
	Rendering Mode: Shaded plus wireframe
	Rendering Mode: Wireframe
	Rendering Mode: Shaded
	Toggle Camera Type: Perspective view
	Toggle Camera Type: Orthographic view
	Show/hide orientation.
	Show gridlines.
	Place a coordinate.
	Place a reference coordinate.
	Perform calculations.
	Change display settings.
	Accept all changes and import them into MapleSim.

Figure 32: The toolbar of the **Import CAD** toolbox.

7 Creating a model using CAD data

In this example, an overhead bridge crane is set up using MapleSim. The procedure is divided into the following parts:

- 1) Create model for the overhead bridge crane.
- 2) Animate the overhead bridge crane.
- 3) Export FMU from the model.
- 4) Integrate the overhead bridge crane model in B&R Automation Studio.

7.1 Generating a library from the CAD data

7.1.1 Importing CAD data

The overhead bridge crane model is made available as a .step file (see [Fig. 33](#)).

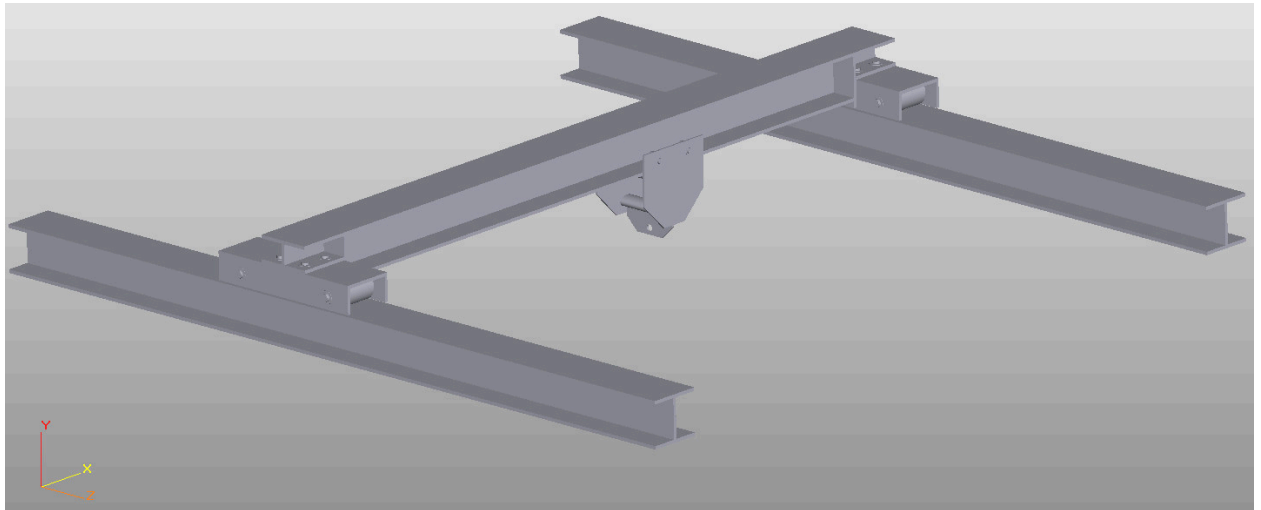


Figure 33: The .step model for the overhead bridge crane.

Defining subgroups

Subgroups of the base and the bridge are then formed (see [Fig. 34](#)).

Each of these subgroups later becomes an object or function block in the visual editor. Individual objects at the highest level are also converted to a function block. In [Fig. 34](#), this is the case for the trolley.

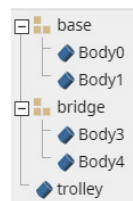


Figure 34: The subgroups of the overhead bridge crane model.

7.1.2 Determining the basic orientation

Creating and assigning coordinate systems

After the CAD data has been imported, each of the subgroups needs its own coordinate system (flange) in order to connect the objects later in the workspace. At the position of the coordinate system there is a flange in the workspace that can be used to connect the objects. To add a coordinate system (flange), click on the corresponding icon according to Fig. 32 and select a point as shown in Fig. 35. Correctly creating the coordinate systems is important for the proper functionality of the entire simulation.

The coordinate systems, as described in Fig. 36, Fig. 38 and Fig. 39 also have to be created.

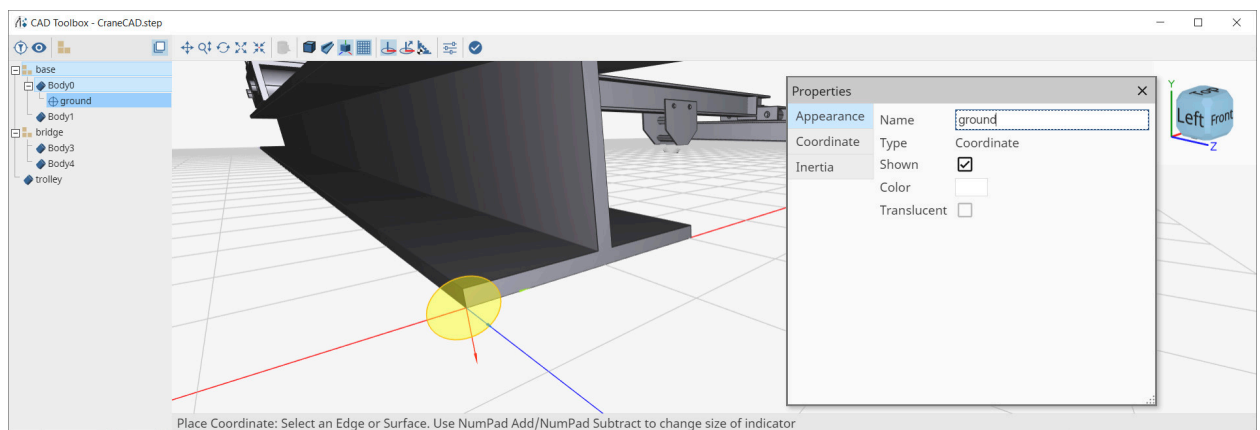


Figure 35: The coordinate system for the base.

Fig. 36 Shows the coordinate system for modeling the connection between bridge and base. In MapleSim Workspace, for example, a prismatic joint can be inserted along the z-axis after import, which can cause the bridge to move along the z-axis relative to the base. In Fig. 36, coordinate system A was added for the bridge. A reference to coordinate system A was added for the base (copy/paste). As a result, the base gets a coordinate system (reference to A) that has the same global coordinates as A, which is assigned to the bridge.

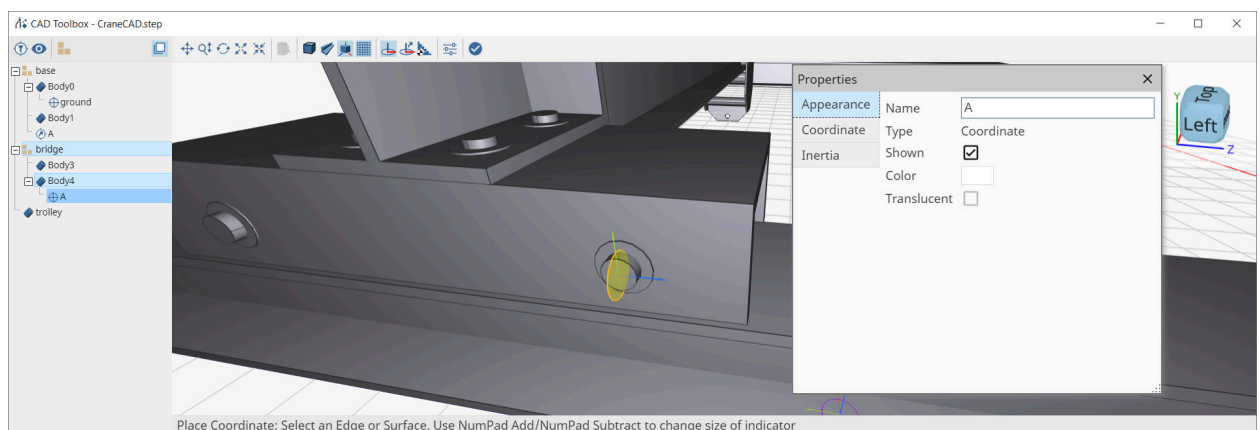


Figure 36: The coordinate systems between base and bridge.

The coordinate system and its reference can be connected in the workspace using a prismatic joint, for example (only allows translation). In Fig. 37, a prismatic joint that allows movement in the z-direction is used ([0,0,1]).

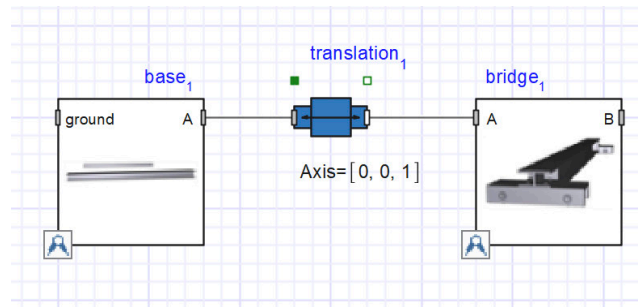


Figure 37: Connection of base and bridge in workspace.

In [Fig. 38](#) (bridge - trolley) and [Fig. 39](#) (trolley - rope), other required coordinate systems are shown.

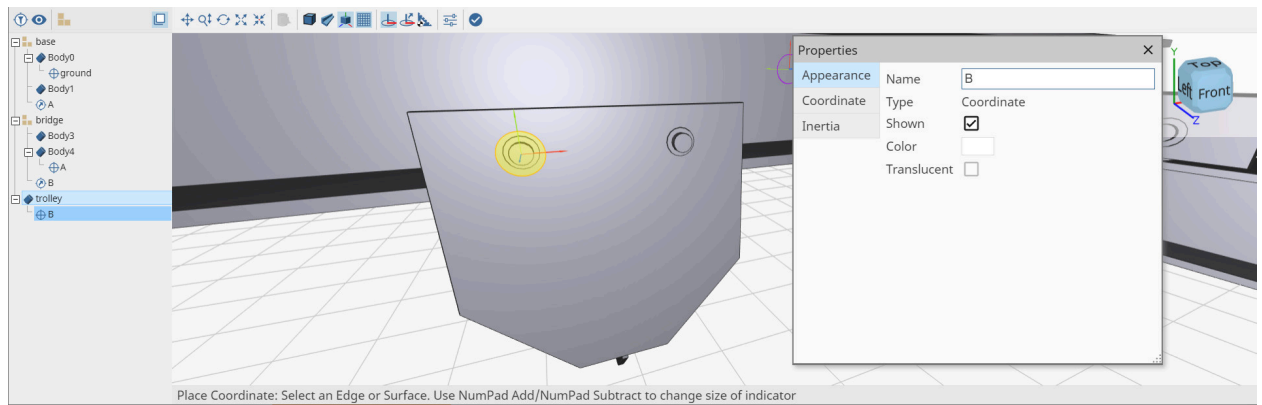


Figure 38: The coordinate systems between bridge and trolley.

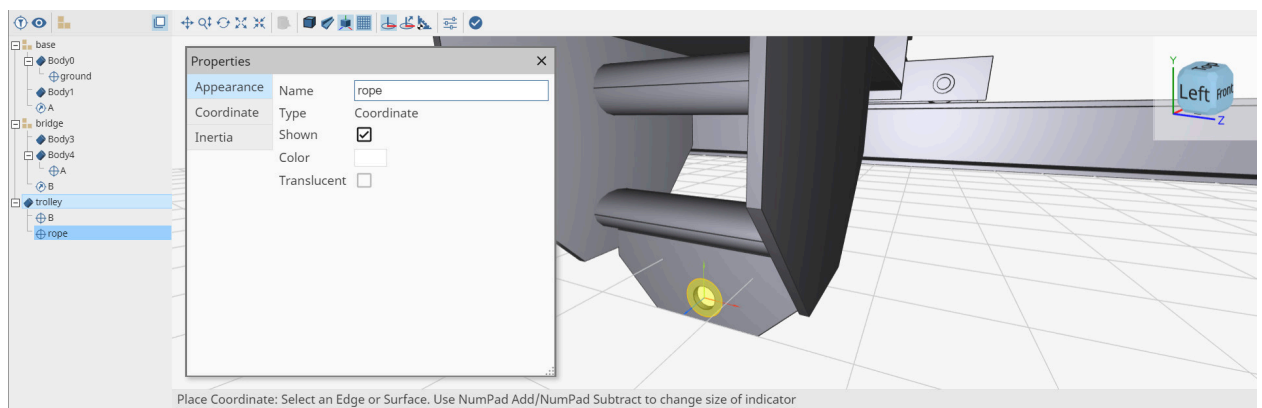


Figure 39: The coordinate systems between trolley and rope.

The **Import** icon  must be pressed to add the parts. After that the parts are in the **Local Components** menu (see [Fig. 40](#)). They can now be used in the workspace.

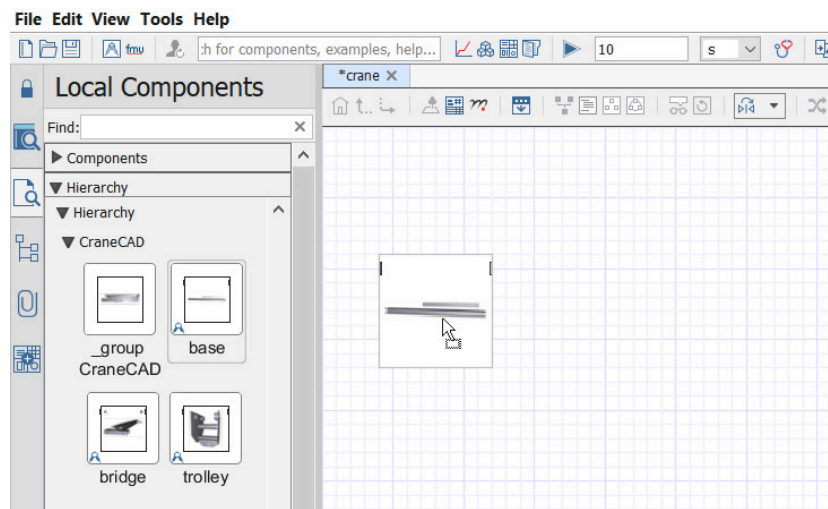


Figure 40: The Local components toolbox.



When importing CAD data, the material properties (right click) should be checked to ensure that they are available and correct. This has a significant effect on the physical behavior of the system.



The coordinate systems and the material properties can also be changed afterwards by double-clicking on the object.

Exercise: Import CAD data

Import the .step file for the overhead crane. Proceed as described in the previous chapter. The following steps are now performed one after the other:

- Import the .step file.
- Define the **groups**. These are
 - Base,
 - Bridge
- Define and assign the coordinate systems.
- Apply the settings.

7.1.3 Identify and define the elements / joints

Add all necessary elements into the model

To be able to add all necessary elements, the imported model must first be checked and then the base must be connected to the **ground**. A **fixed frame**, as described in [Fig. 41](#) is used (other elements, which are important for the example, are also highlighted in the image).

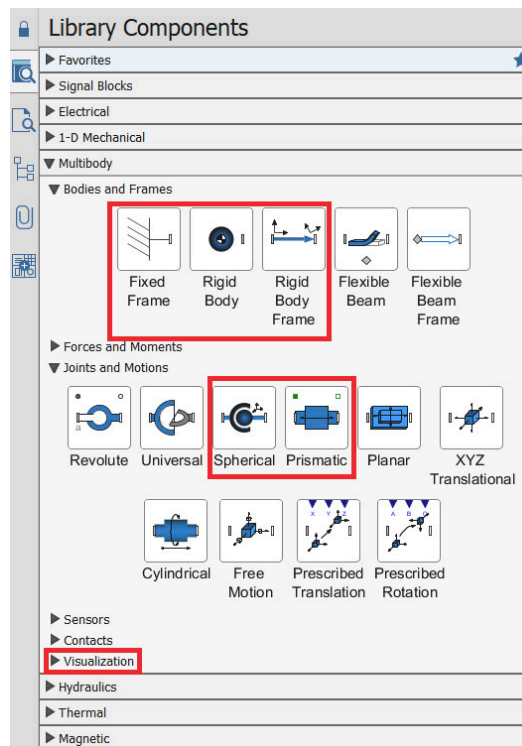


Figure 41: Elements in the Multibody toolbox.

Now that the CAD data has been imported, the individual parts (base, bridge and trolley) must be joined together with joints. Important types of joints for this example are **spherical** and **prismatic**, see Fig. 41. **Spherical** corresponds to an element rotating around an axis, whereas **prismatic** corresponds to a sliding element.

Fig. 42 shows an excerpt from the MapleSim help documentation related to a **prismatic joint**.

The cube (2) can only move along one axis (here the x-axis). Two coordinate systems are needed to generally describe the relative motion between the bearing (1) and the cube (2). One coordinate system for the bearing (1) and one coordinate system for the cube (2).

With the help of the coordinate systems, it is possible to carry out a rotational or translational movement of an object in all directions. If the trolley is connected to the bridge in this way (with the help of a **prismatic joint**), the bridge can be moved on the base along the x-axis.



Two coordinate systems are always required for a relative movement between two objects. This is achieved simply by forming a coordinate system for the first body, which is then copied (or referenced) for the second body and adapted accordingly.

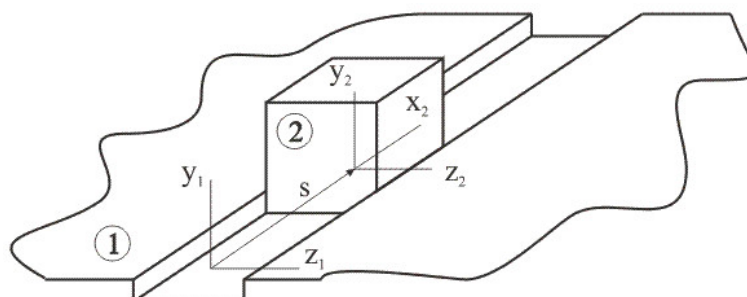


Figure 42: The prismatic joint.

Exercise: Creating an initial model with CAD elements

Use a **fixed frame**, a **spherical joint** and the base to create a model (see [Fig. 43](#)). If you now simulate/run the model, what results can you see in the **3D playback** window?

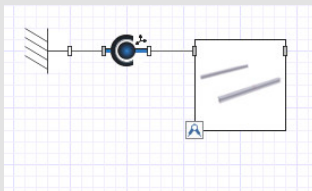


Figure 43: An initial model with a self-made subgroup.



Check the tooltip (hover over the connector with the mouse) on the base to determine if you are connecting the correct input.



To test the model, the play icon  can be pressed in the Main toolbar. To switch to the **3D playback window**, the corresponding icon can be clicked . The movement of the model can be examined and verified here (**3D playback** window - see e.g. [Fig. 46](#)).

Coupling the elements

Now the complete model for the overhead bridge crane will be created. For reasons of simplicity, the model is built in an initial step with a fixed rope length. The rope is modeled using a rod, which is an approximation of reality.

To link the elements, the connections are set up as described in [Fig. 44](#). The properties must be set in accordance with [Fig. 45](#).

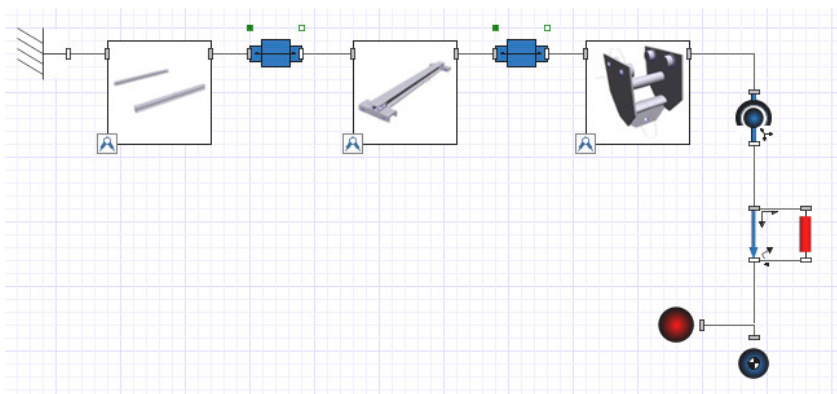


Figure 44: Model of the complete overhead bridge crane in the MapleSim workspace.

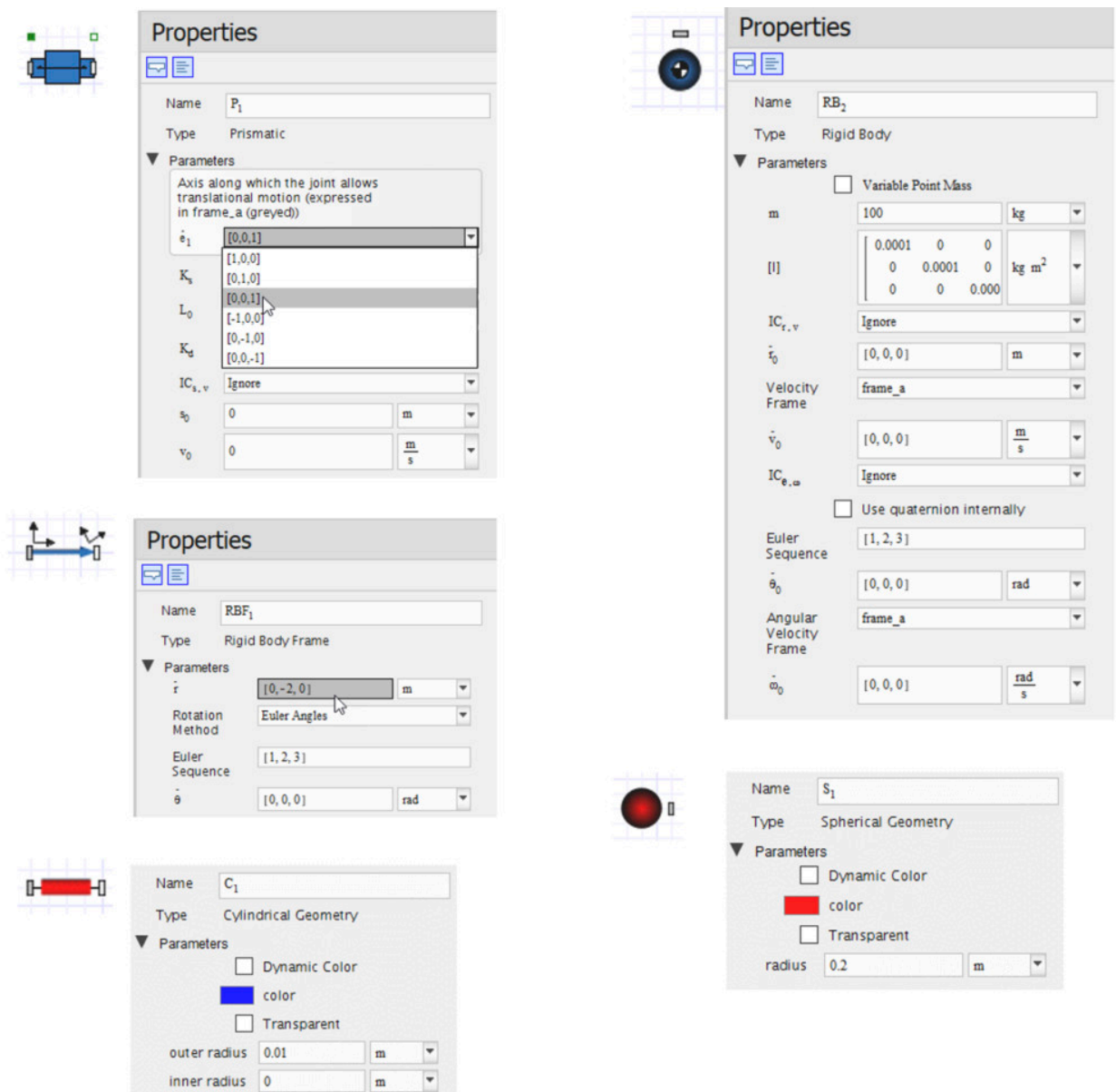


Figure 45: Values of the elements in the model.

Exercise: Creating the complete model

Create the complete model of the overhead bridge crane as described in Fig. 44 and test it.

- The function blocks created from the CAD data are added to a new model in combination with elements from the **Multibody** library.
- These elements are then coupled together.
- The parameters of the **visualization** elements for **cylindrical geometry** and **spherical geometry** must be adapted to obtain a realistic visualization of rope and mass.

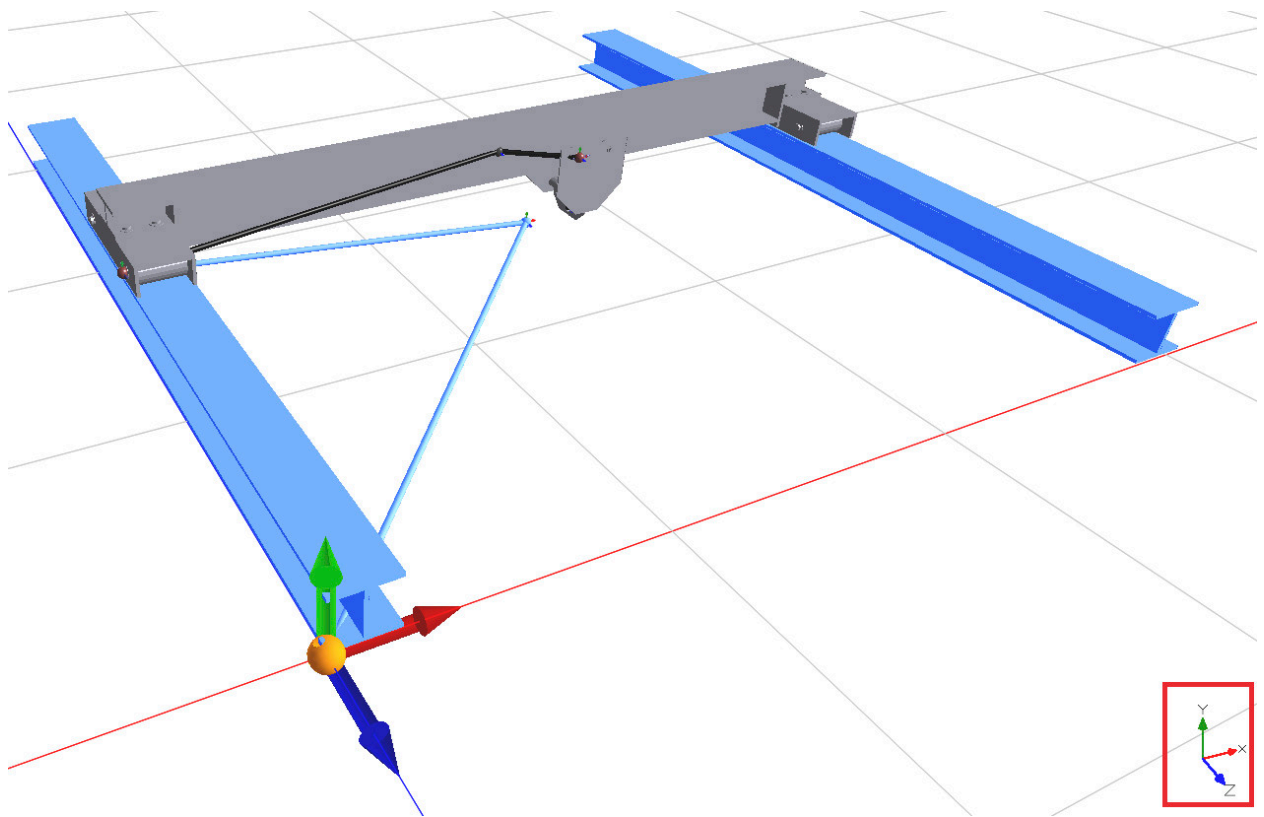


Figure 46: Model of the overhead bridge crane visualized in **3D playback**.

7.1.4 Installation of 1D elements

In order to animate the model, elements from the **1D mechanical** library are required.

When animating the model with the aid of trajectories, the following steps must be carried out:

- The drive/trajectory of the trolley must be defined.
- A gearbox for the bridge must be installed.
- The drive/trajectory for the bridge must be defined.
- Measurement sensors must be set.

The following software modules are used in this chapter:

- **1D Mechanical library**
- **1D Motion Generation**
- **B&R SERVOfsoft data generation**

Connection to the 1D mechanical domain

Some joints offer a connection to the **1D mechanical** library. For example, the previously used **prismatic joint**  and **revolute joint** . These joints have connections (framed in red in the small pictures) that can be connected to a drive, a sensor or other mechanical systems. These connections are bidirectional.

Adding a gearbox to the bridge

At this point, a gearbox is added to the model. This allows the translatory bridge to be connected to a rotary drive. In the following images, the required components are shown.

The source used is a sinusoidal signal whose output must be converted into a translational position unit in order to be connected to the flange of the **prismatic joint**. In order to use a rotary unit for a drive (as shown in the example for the trolley), the corresponding rotary **motion driver** must be used.

The desired value for the gear ratio is set for the gearbox (see Fig. 47).

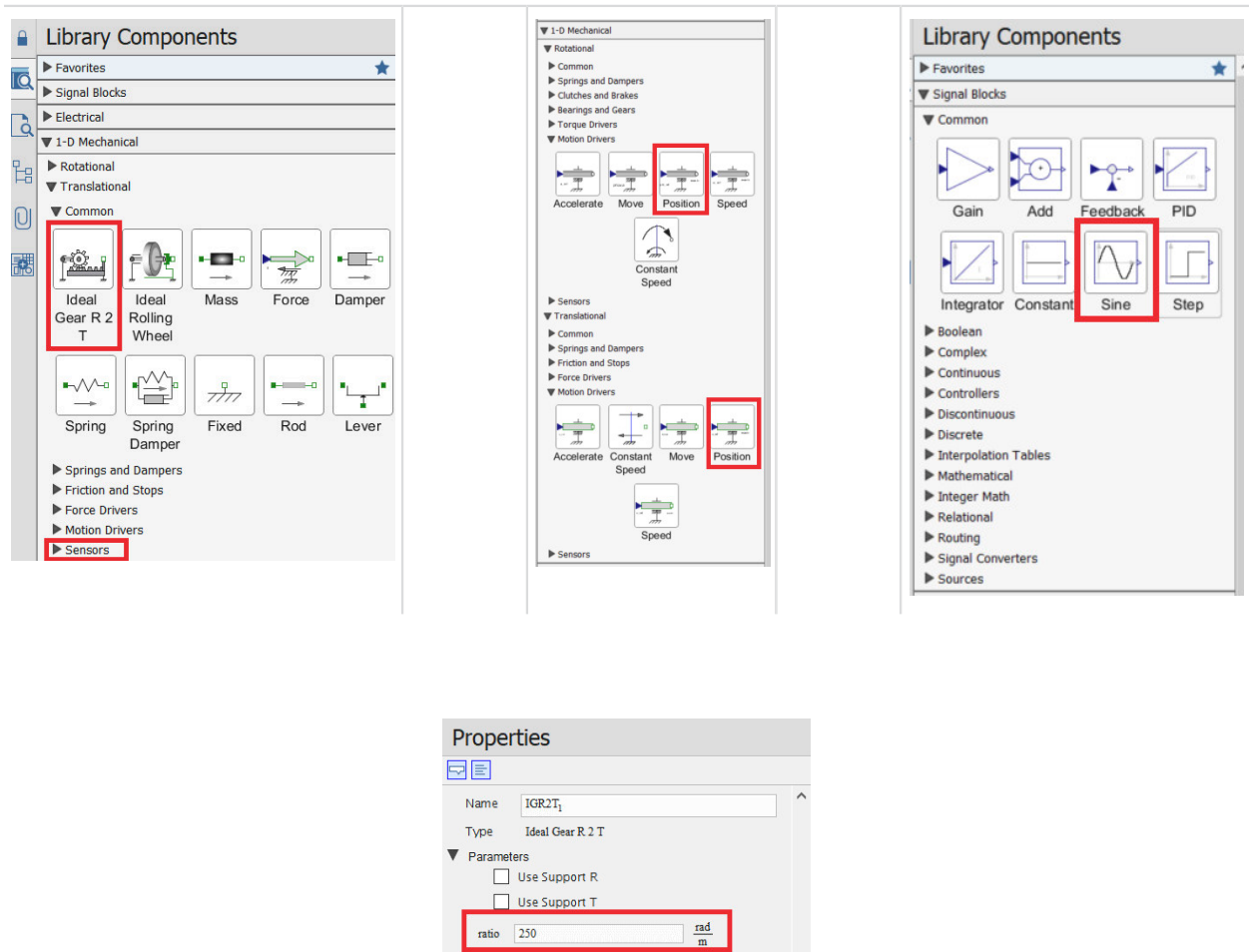


Figure 47: MapleSim elements for modeling, as well as the gear ratio parameters.

Exercise: Installing 1D elements in the model

Add the corresponding 1D elements to the model.

- Use a rotary drive for the bridge element. Tip: Use a gearbox.
- Use a translational drive for the trolley.

7.2 Subsystems

Subsystem of the overhead bridge crane

The subsystem of the overhead bridge crane is created as follows (see Fig. 48).

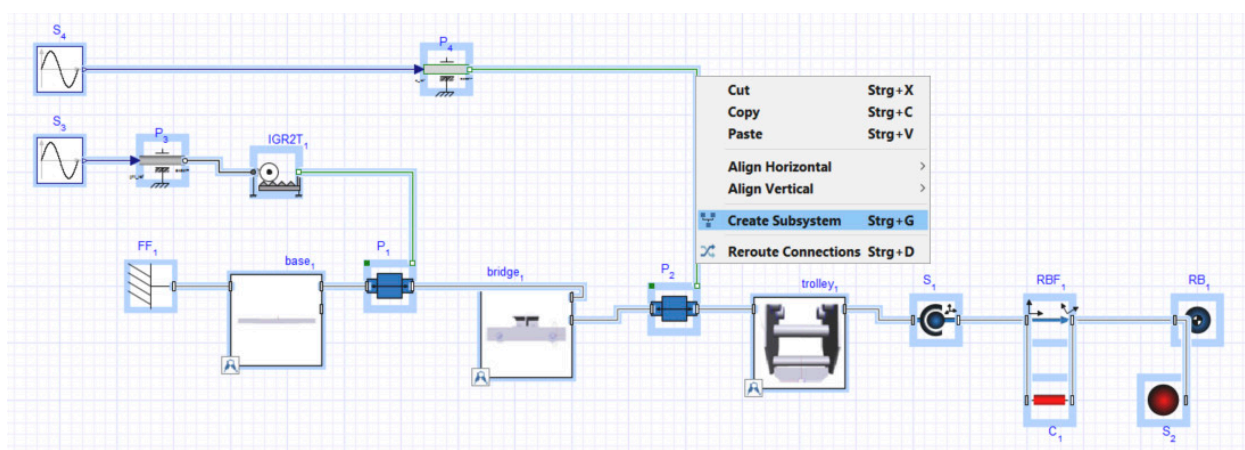


Figure 48: Create a subsystem using the menu accessed with the right mouse button.

After creating the subsystem, the bidirectional inputs must be replaced with unidirectional ones because the interface of the FMU only supports **LREAL** (double) types.

Use of a motion driver based on speed

The **position drives** are replaced by **speed drives** (Fig. 49 and Fig. 50). This makes it easy to specify a movement in the Watch Window later in Automation Studio, which simplifies testing. If you use a position drive and then later set a large position jump in the Watch Window, for example, the simulation would crash because position jumps - even in reality - are not possible.

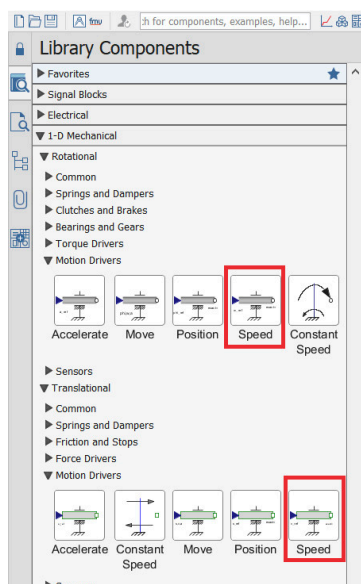


Figure 49: Drives for the mechanics - Domains.

In MapleSim, unidirectional inputs of subsystems are marked with arrows. Functions and function blocks in B&R Automation Studio can only process unidirectional inputs. As mentioned, only **LREAL** variables are permitted when importing FMUs into B&R Automation Studio.

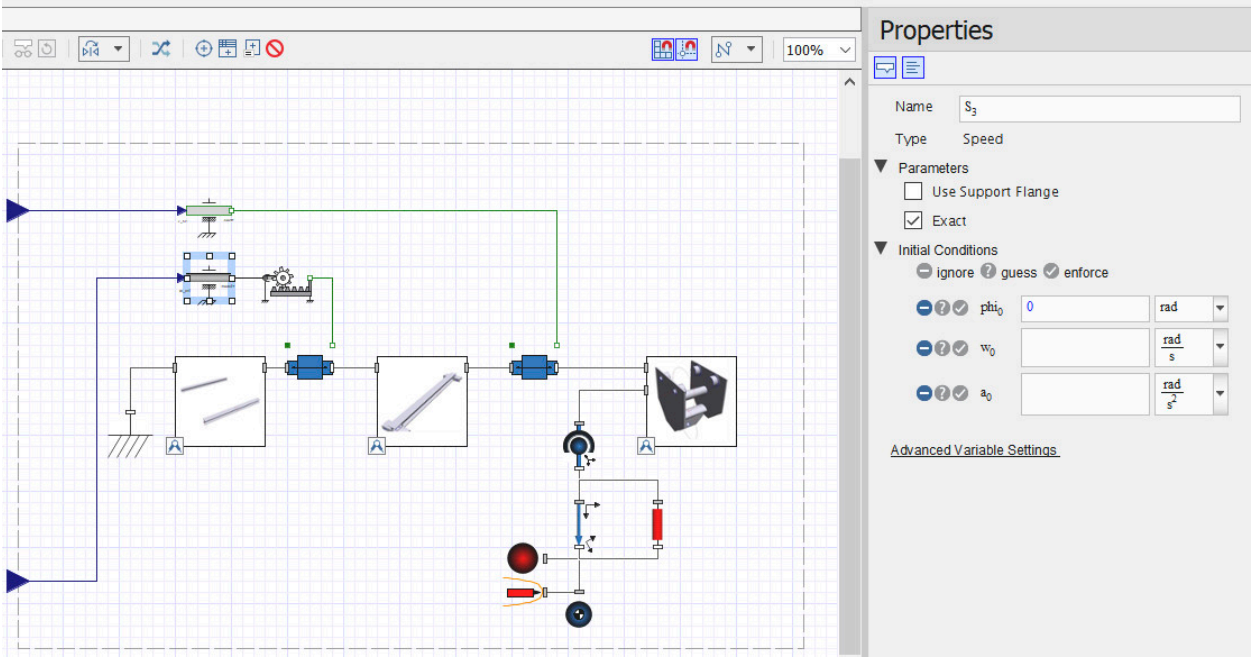


Figure 50: Subsystem for the overhead bridge crane.

Using the fixed step solver

The solver of the model must be set to a **fixed step solver** (see Fig. 51). The step size must correspond to the cycle time in B&R Automation Studio.

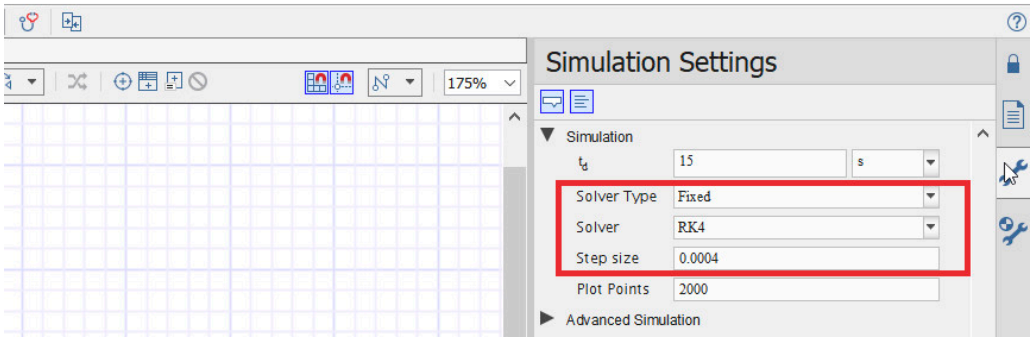


Figure 51: The FMU model should run with a **fixed step solver** (400 μ s).

The MapleSim help documentation offers support for simulation settings. The following table shows a description of different solvers in MapleSim.

Solver type	Variable	The type of solver used for the simulation. - Variable: A variable step size is selected to keep the error tolerance low. - Fixed: Selecting a fixed step size does not take the error tolerance into account. Important: Solvers with a fixed step size are identical to those used by MapleSim in the exported code.
Solver	Variable: CK45 (non-rigid) Fixed: Euler	These are DAE solvers that are used during simulation. The following options are available for a variable step size: - CK45 (semi-rigid): - RK45 (non-rigid): - Rosenbrock (rigid): If the model is complex, a rigid DAE solver can be used to minimize the time necessary to simulate. The following options are available for solvers with a fixed step size: - Euler: Forward Euler solver - Implicit Euler: An implicit Euler solver, also suitable for rigid systems - RK2: Second-order Runge-Kutta solver - RK3: Third-order Runge-Kutta solver - RK4: Fourth-order Runge-Kutta solver

Different solver options in MapleSim.

Exercise: Step size and instability

Check if the model is running stable with the selected step size. In addition, the system should be tested to determine the step size at which the model becomes unstable. Why does the system become unstable?

- Embed the overhead bridge crane model in a subsystem.
- Swap out the **motion driver** elements related to position for those related to speed.
- Select a fixed step size with an RK4 solver and test the system.
- Try out how high the step size can be set without making the system unstable.
- Explain why the system can become unstable.

7.3 Recording signals in the model

Torque output of the bridge controller

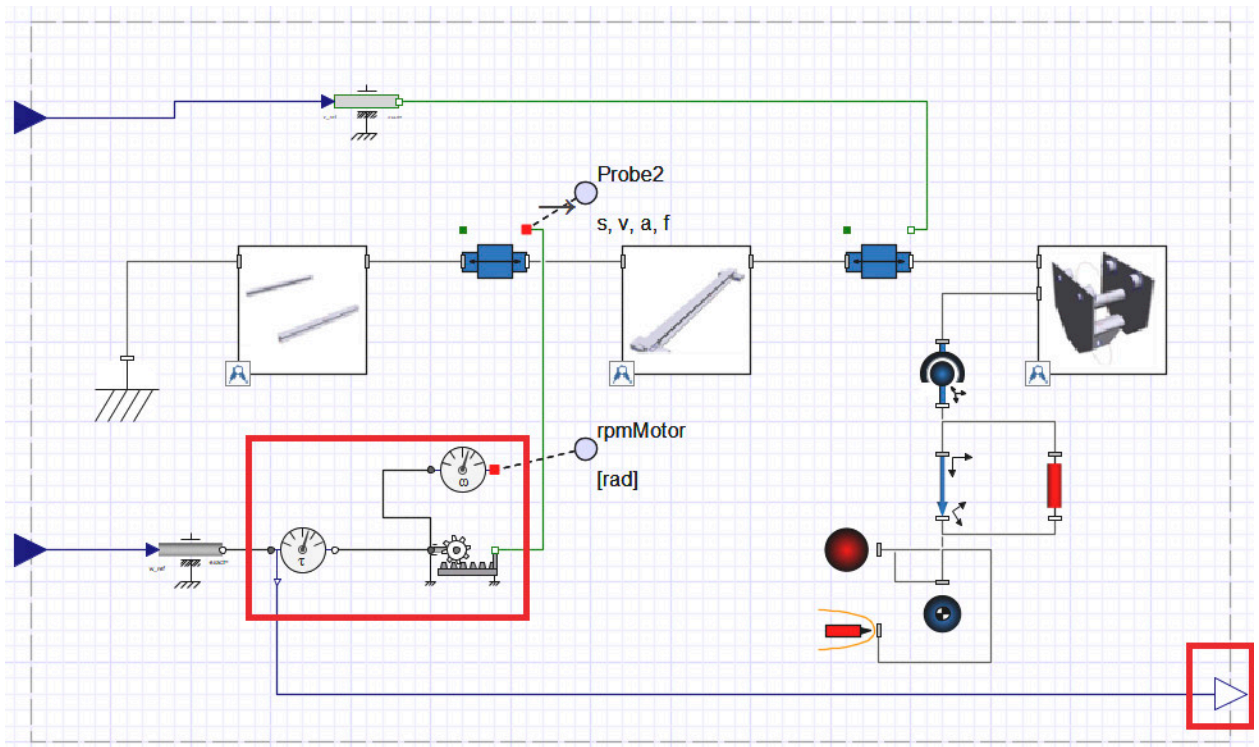


Figure 52: Add a torque and speed sensor to the model.

The torque must be output via a connection between the sensor and the outer boundary of the subsystem (see Fig. 52). The measurement sensor must be coupled to the model with an analog connection Fig. 53.

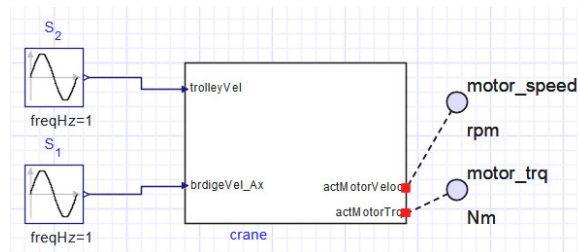


Figure 53: The crane model, ready for FMU export.

7.4 Setting user-defined input values

It is possible to generate user-defined signals in MapleSim that can be used as motion profiles. This can be done under **Add Apps or Templates** by selecting the **Component Creation** area and then **1D Motion Generation**. The motion profiles created in this way can then be selected as blocks under **Local Components** under **Components**. To illustrate this, the first step is to generate a motion profile for the trolley.

Set up the trolley drive

In the 1D Motion Generator, the signal path can be created. First, the various segments of the signal are created. As signal type we will use Signal. In Fig. 54, the required areas are marked in red.

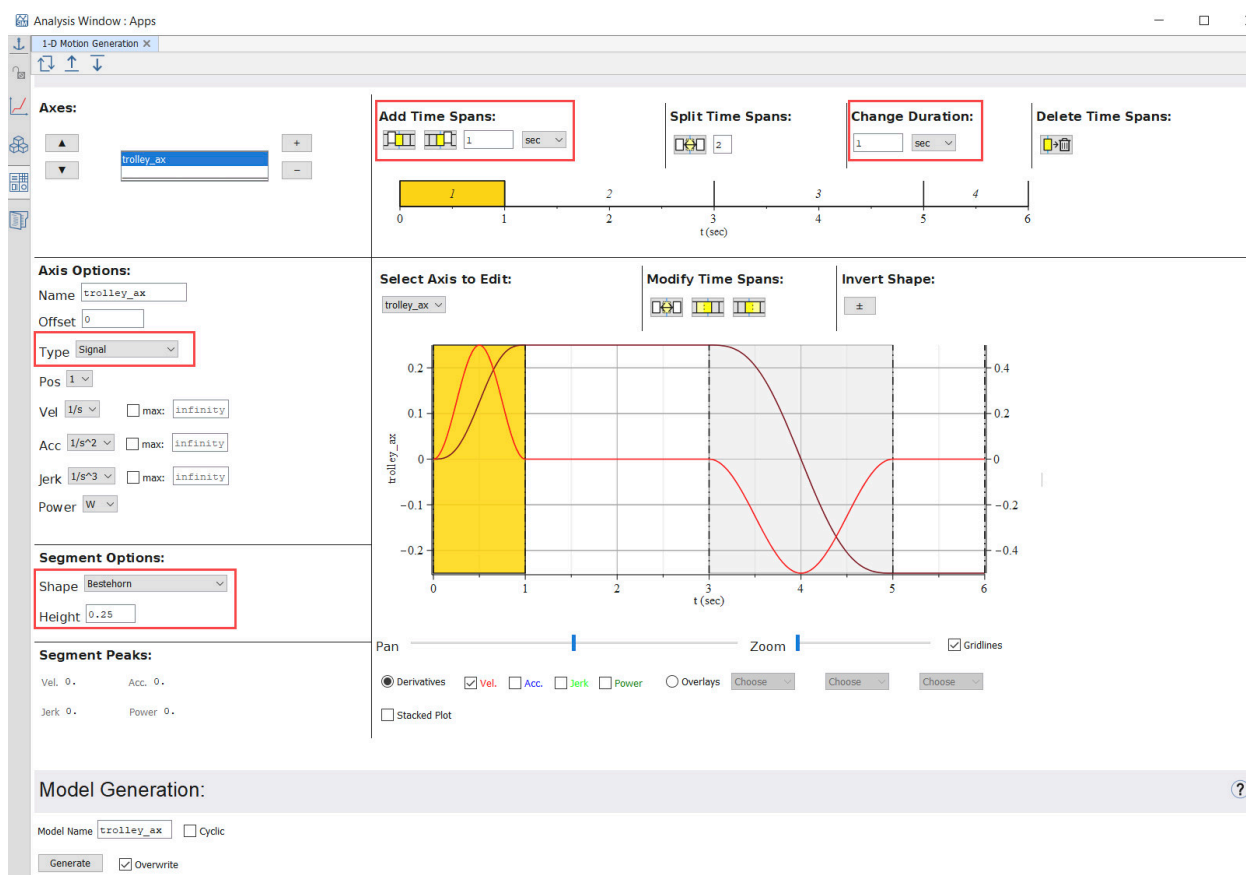


Figure 54: The motion profile for the trolley.

After the motion profile for the trolley has been defined, it must be added as a block and connected to the translational joint between the bridge and the trolley. The block now replaces the Sine block. Fig. 55 shows this connection in the workspace.

Connecting multi-dimensional signals

If Signal is selected as the signal type when creating a motion profile with the 1D Motion Generation app, a 3D signal is output at the block output. The signal consists of position, speed and acceleration. In our example we only need the speed, so we select the 2nd signal (Fig. 55)

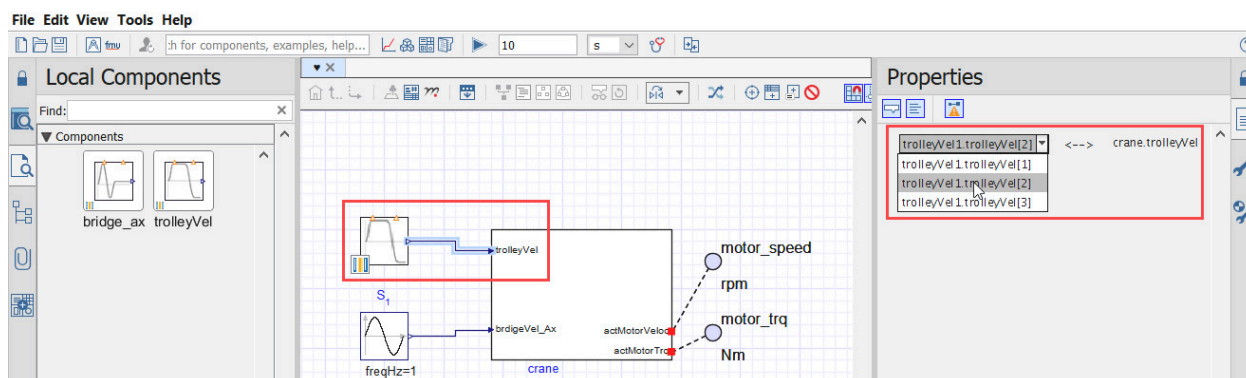


Figure 55: Connecting the speed from the 3 dimensional output of the motion profile

Set up the drive for the bridge

The motion profile for the bridge drive is now defined, as shown in Fig. 56.

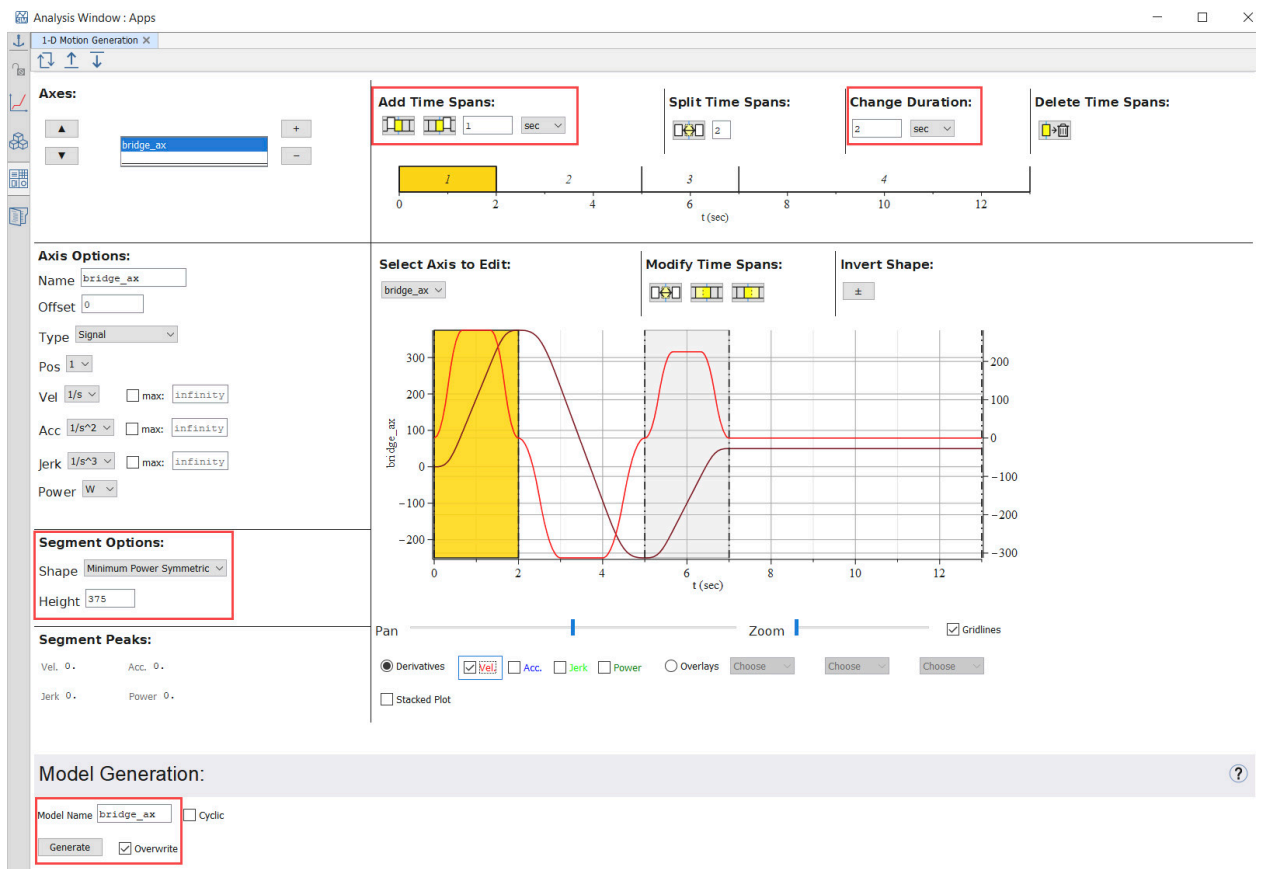


Figure 56: The motion profile for the bridge motor.

Limits to motor speed can be calculated:

$$v_{\max} = 280 [\text{rad/s}] = 2675 [\text{rpm}]$$

A motor rated at 3000 revolutions per minute can therefore be selected. Converting between the units is done as follows:

$$[\text{rad/s}] = 2\pi / 60 [\text{rpm}] = \pi / 30 [\text{rpm}]$$

Exercise: Motion profiles

Create corresponding motion profiles using the **1D Motion Generator** and connect it to the model for the overhead bridge crane.

- 1) Create the motion profile for the trolley and the bridge using the **1D Motion Generator**.
- 2) Connect these trajectories to the subsystem and test it.

7.5 Verification of the FMU

Adding measurement sensors

Traces can be recorded in MapleSim using measurement probes that have been added to the model. This allows, among other things, position, speed, acceleration and force to be measured. The results can be checked after running the simulation based on Fig. 57 and Fig. 58.

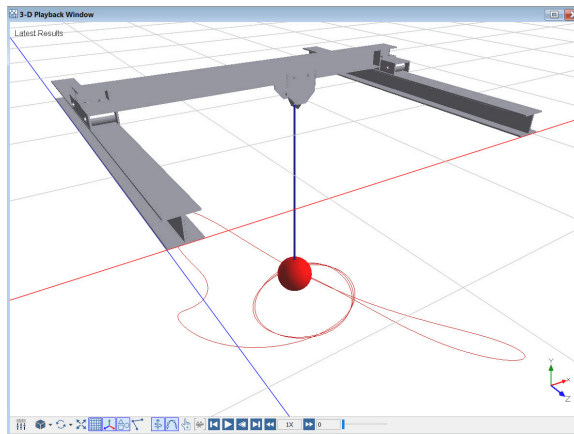


Figure 57: The complete model of the overhead bridge crane.

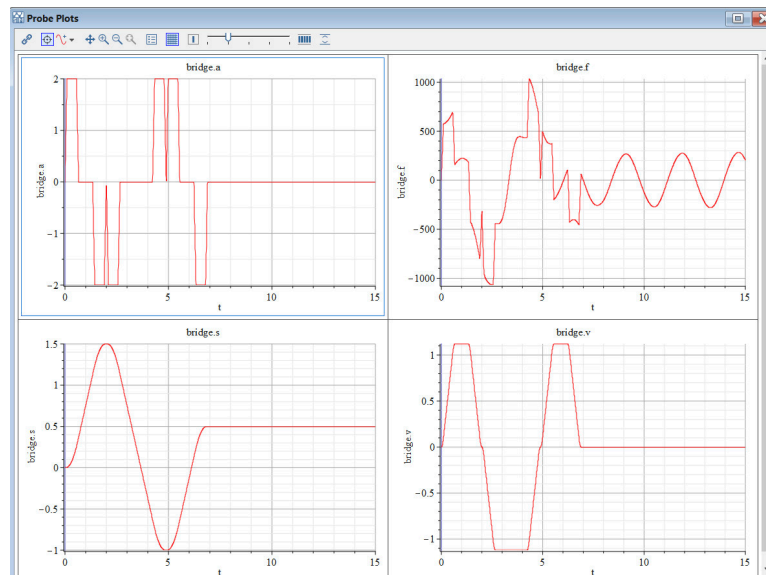


Figure 58: Results of the complete overhead bridge crane.

Exercise: Testing the model - Installing the measurement probes.

- Add the measurement sensor (**probe**) for the overhead bridge crane model. Decide for yourself which parameters need to be included to verify the model you have created.
- Start the simulation and record the measured values.
- Try using the measured values to determine if the model is correct and why.

7.6 Exporting the Functional Mock-up Unit for the overhead bridge crane

The next step is opening the FMU export to be able to export the model (see [Fig. 59](#)). This can be found in the tab **Add apps or templates / B&R connector**.

The corresponding parameters, which are important for proper export, can be found here: [Fig. 59](#).

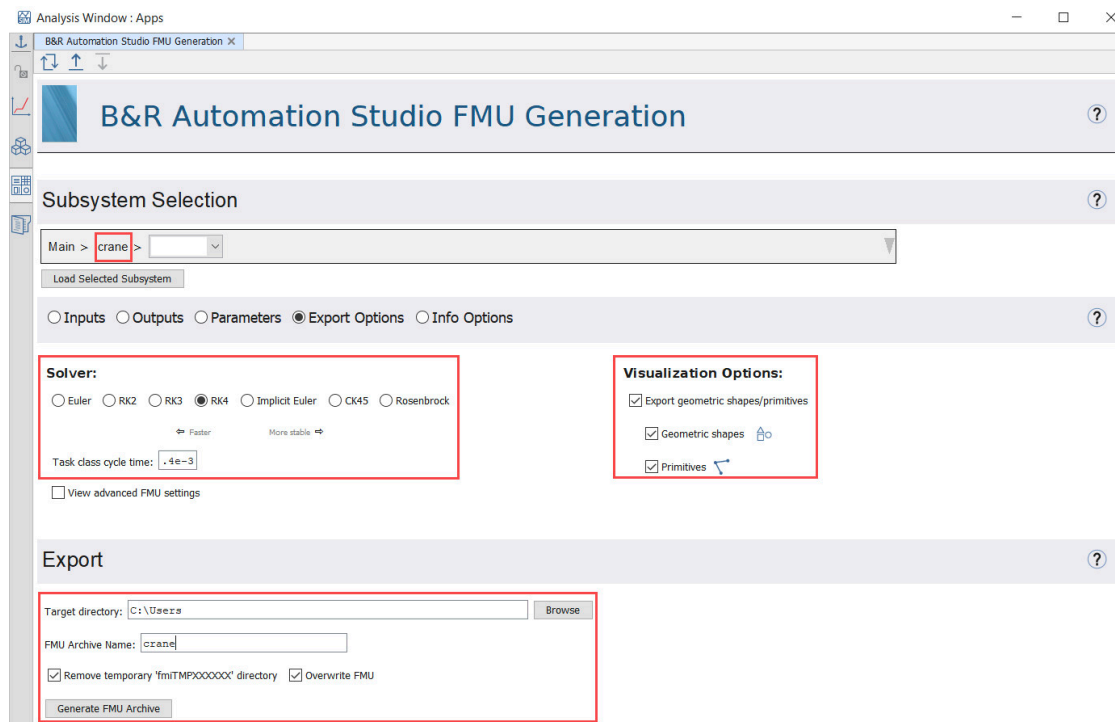


Figure 59: Parameters for FMU export of the overhead bridge crane.

Exercise: Exporting the model with the B&R Automation Studio FMU Generator

- Open the FMU generator.
- Set the parameters for proper export.
- Export the FMU.

7.7 Integration in a B&R Automation Studio project

Visualization with B&R Scene Viewer

One of the first steps in integrating the FMU is to open B&R Automation Studio and create a new project. The FMU can then be installed in **Toolbox - Object catalog -> Library -> FMU library** (see Fig. 60).

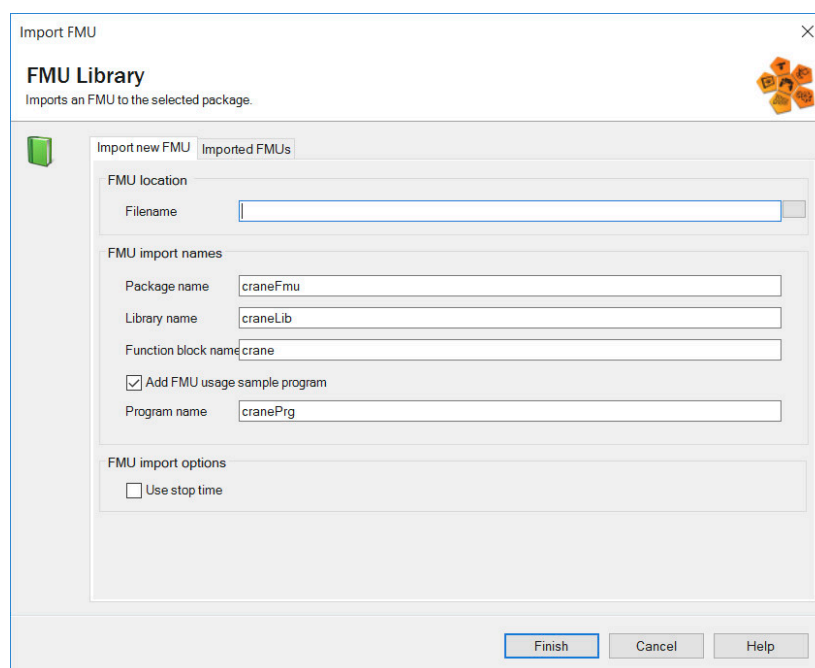


Figure 60: Importing the overhead bridge crane FMU into B&R Automation Studio.

The program must be assigned to the corresponding **Cyclic#** with the correct sampling time. It is important that the sampling time matches the cycle time of the FMU from MapleSim (in this case, this is a **fixed step solver** with 400 μ s sampling time (see [Fig. 61](#))).



It is important to note that the higher the complexity of the system and the lower the sampling time, the more computing power the computer/PLC must provide. It can therefore often be useful to select a long sampling time to keep the computing power low. The sampling time cannot be excessively long, however; otherwise, solving equations with the solver will become unstable. A reasonable compromise must be found here.

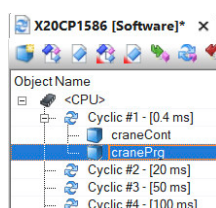


Figure 61: Assigning the tasks in the Cyclic#.

B&R Scene Viewer can now be opened by double-clicking on the corresponding icon (see [Fig. 62](#)).

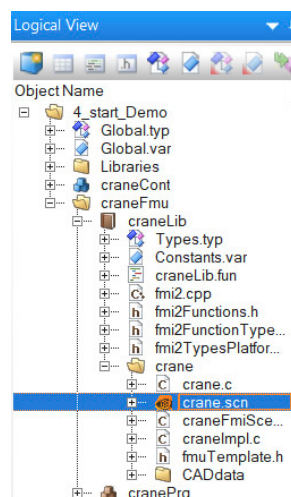


Figure 62: Opening the B&R Scene Viewer from B&R Automation Studio.

It is necessary to check if the PVI settings in the Scene Viewer are correct (see [Fig. 63](#)), then the connection to ARSim/AR can be established.

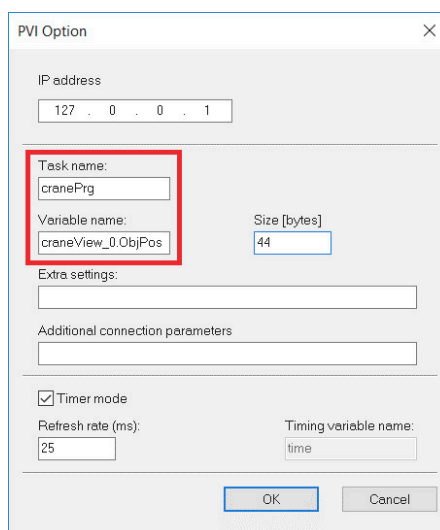


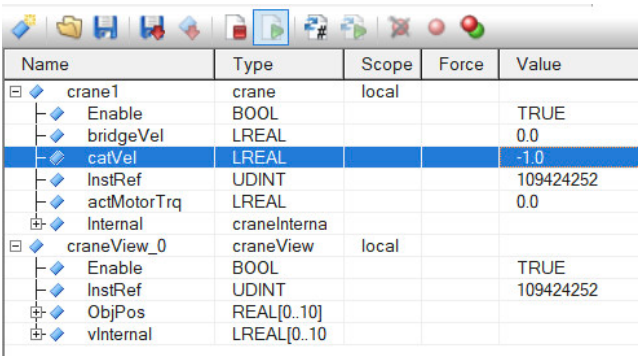
Figure 63: Values for the PVI settings in B&R Scene Viewer.

Now the model can be controlled via the Watch window (see [Fig. 64](#)).



It is important to set both ENABLEs in the Watch window to TRUE so that the connection to the B&R Scene Viewer is enabled and the visualization is possible (see [Fig. 64](#), crane1.Enable = TRUE, craneView_0.Enable = TRUE).

In the <model_name>View_0 function block, the calculations for visual representation in B&R Scene Viewer are performed. If visualization is not needed, it can be switched off, thus saving computing power.



Name	Type	Scope	Force	Value
crane1	crane	local		
Enable	BOOL			TRUE
bridgeVel	LREAL			0.0
catVel	LREAL			-1.0
InstRef	UDINT			109424252
actMotorTrq	LREAL			0.0
Internal	craneInternal			
craneView_0	craneView	local		
Enable	BOOL			TRUE
InstRef	UDINT			109424252
ObjPos	REAL[0..10]			
vInternal	LREAL[0..10]			

Figure 64: The Watch window in B&R Automation Studio for controlling the B&R Scene Viewer.

Exercise: Integration in B&R Automation Studio

Add the FMU to a new B&R Automation Studio project and test it.

- Import the FMU into B&R Automation Studio as an FMU library.
- Open B&R Scene Viewer.
- Establish a connection to the B&R Scene Viewer.
- Set the corresponding Enable values to TRUE.
- Test the program. You should now be able to maneuver the crane.

8 SERVOfsoft toolbox

SERVOfsoft can be used to size drives. The **B&R SERVOfsoft Data Generation** toolbox in MapleSim is used to generate the necessary data for this purpose (see Fig. 65). From here, the **B&R SERVOfsoft Data Generation** toolbox obtains the necessary dynamic information to be able to size a motor.

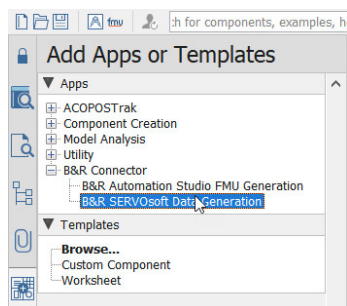


Figure 65: The data generator from SERVOfsoft.

After the toolbox has been launched, the system can be loaded (see Fig. 66). In order to configure the drive for the bridge motor, the bridge must be specified under **actuator selection**. In addition, **rpm** must be selected under **Output units** for the speed. The profile can be displayed by pressing the "Preview export" button.

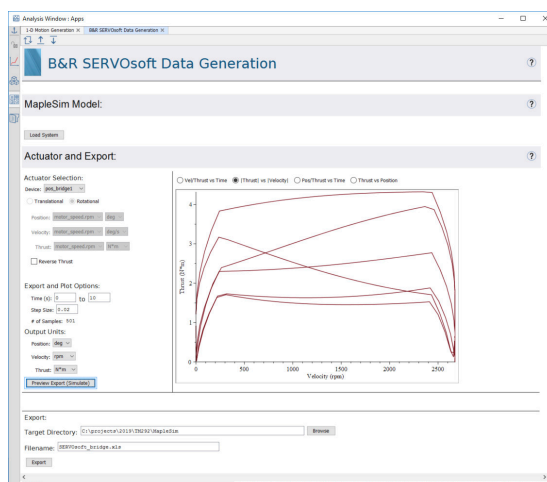


Figure 66: The chart of a velocity/torque curve generated by the motion generator.

An .xls file can then be generated by selecting a suitable directory and file name and pressing **Export**. The file created contains the sampling time, the rotation speed, the inertia (always 0 in the MapleSim export) and the torque. This data can now be loaded from SERVOfsoft once the **(Rotary)** load is selected and the **Sequence** button is clicked (see Fig. 67).

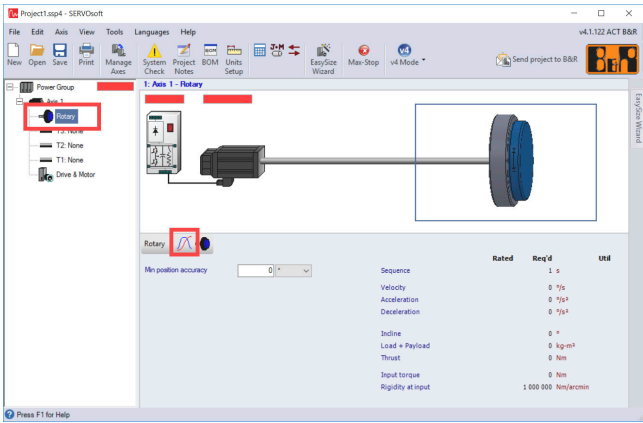


Figure 67: SERVOSOFT - Loading the load profile.

In the newly opened wizard, the **import** function must be selected on the Sequence tab (see Fig. 68).

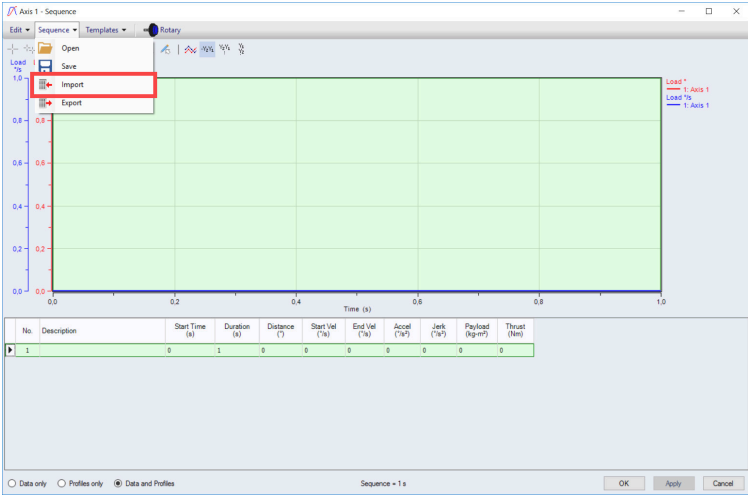


Figure 68: SERVOSOFT - Importing data from the .xls file.

In the following window, **Time and velocity** and **Payload and thrust** must now be selected (see Fig. 69).

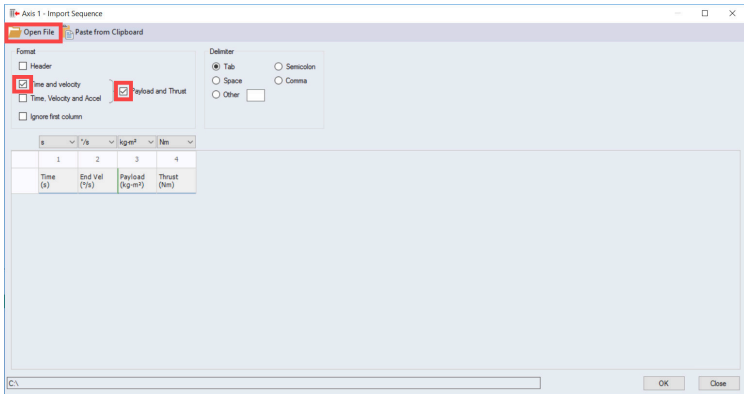
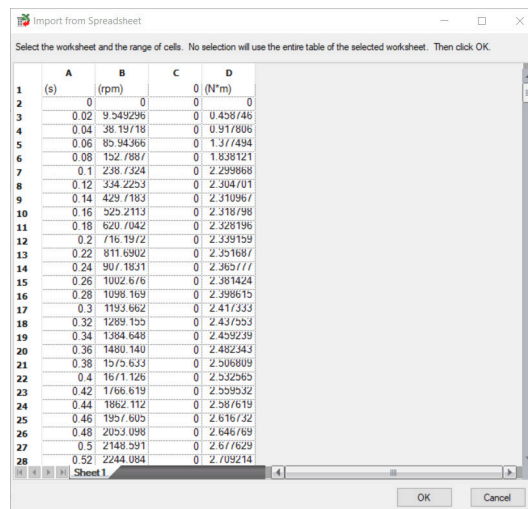


Figure 69: SERVOSOFT - Selecting the data to be imported.

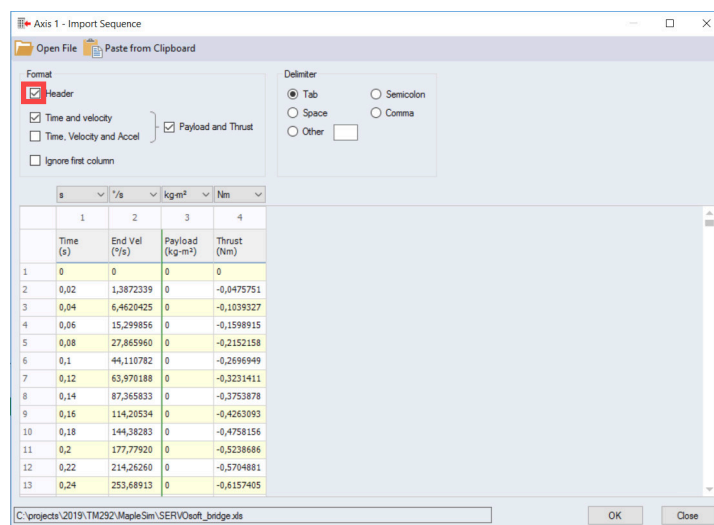
The data is now displayed in a newly opened wizard (see Fig. 70).



	A	B	C	D
	(s)	(rpm)		(N*m)
1	0	0	0	0
2	0.02	9.249296	0	0.458746
3	0.04	38.19718	0	0.917806
4	0.06	85.94366	0	1.377494
5	0.08	152.7887	0	1.838121
6	0.1	238.7324	0	2.299868
7	0.12	334.2253	0	2.364701
8	0.14	429.7183	0	2.310967
9	0.16	525.2113	0	2.318796
10	0.18	620.7042	0	2.328196
11	0.2	716.1972	0	2.339159
12	0.22	811.6902	0	2.351687
13	0.24	907.1831	0	2.365777
14	0.26	1002.676	0	2.381424
15	0.28	1098.169	0	2.398615
16	0.3	1193.662	0	2.417333
17	0.32	1289.155	0	2.437553
18	0.34	1384.648	0	2.459239
19	0.36	1480.140	0	2.482343
20	0.38	1575.633	0	2.506809
21	0.4	1671.126	0	2.532565
22	0.42	1766.619	0	2.559532
23	0.44	1862.112	0	2.587619
24	0.46	1957.605	0	2.616732
25	0.48	2053.098	0	2.646769
26	0.5	2148.591	0	2.677629
27	0.52	2244.084	0	2.709214

Figure 70: SERVOsoft - Data from MapleSim.

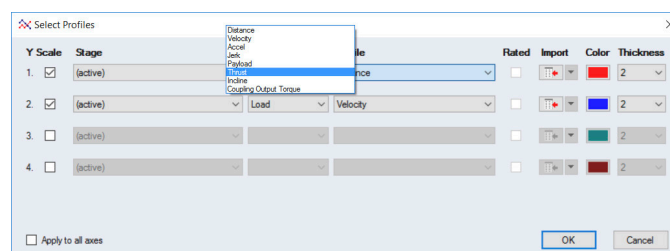
Pressing **Ok** applies the data (see Fig. 71). It can then be seen that the first line is displayed incorrectly. Since there is a header line in the .xls file, it must be eliminated by selecting the **Header** option.



	1	2	3	4
	Time (s)	End Vel (%/s)	Payload (kg-m²)	Thrust (Nm)
1	0	0	0	0
2	0.02	1.3872339	0	-0.0475751
3	0.04	6.4620425	0	-0.1039327
4	0.06	15.299856	0	-0.1598915
5	0.08	27.865960	0	-0.2152158
6	0.1	44.110782	0	-0.2696949
7	0.12	63.970188	0	-0.3231411
8	0.14	87.365833	0	-0.3753878
9	0.16	114.20534	0	-0.4263093
10	0.18	144.38283	0	-0.4758156
11	0.2	177.77920	0	-0.5238686
12	0.22	214.26260	0	-0.5704881
13	0.24	253.68913	0	-0.6157405

Figure 71: SERVOsoft - Removing the header.

Confirm to display the data graphically (see Fig. 73). If the source for the curves should be changed (e.g. **Thrust** - see Fig. 72), this can be done by clicking **Select Profiles** (see Fig. 73).



Y Scale	Stage	Profile	Rated	Import	Color	Thickness
1. [X] (active)	[X] (active)	Distance	[X]	[X]	Red	2
2. [X] (active)	[X] (active)	Velocity	[X]	[X]	Blue	2
3. [] (active)	[] (active)	Acceleration	[]	[]	Green	2
4. [] (active)	[] (active)	Load	[]	[]	Red	2

Figure 72: SERVOsoft - Selecting the profile.

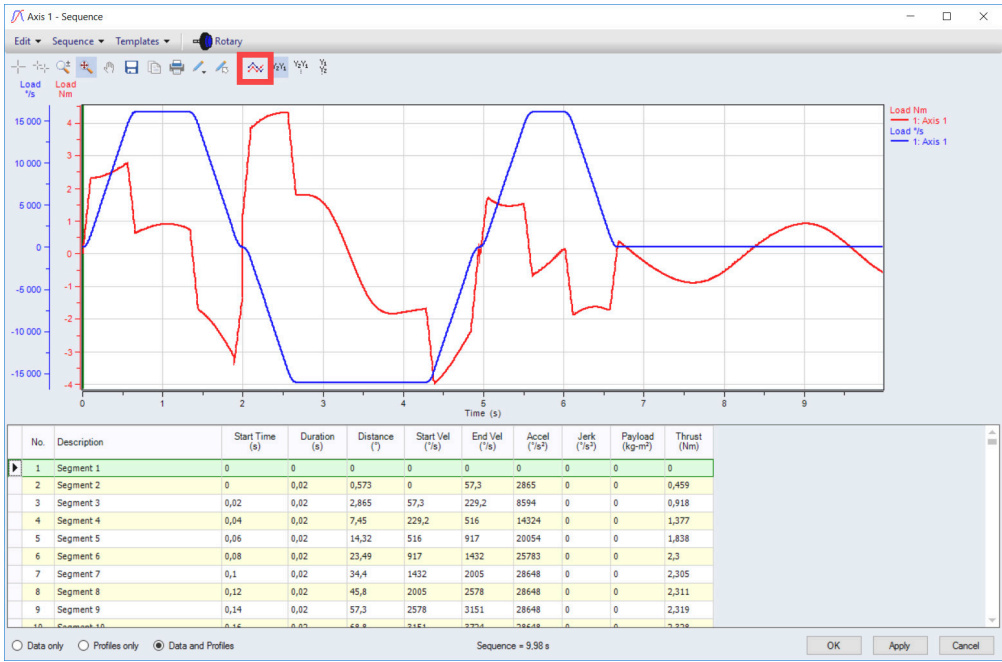


Figure 73: SERVOSOFT - Profiles for speed and thrust.

After confirmation, the load profile is now applied and can be used to size the drive (see Fig. 74). To do this, select the drive unit (**Drive and motor**) and then press the **Open servo motor database** button.

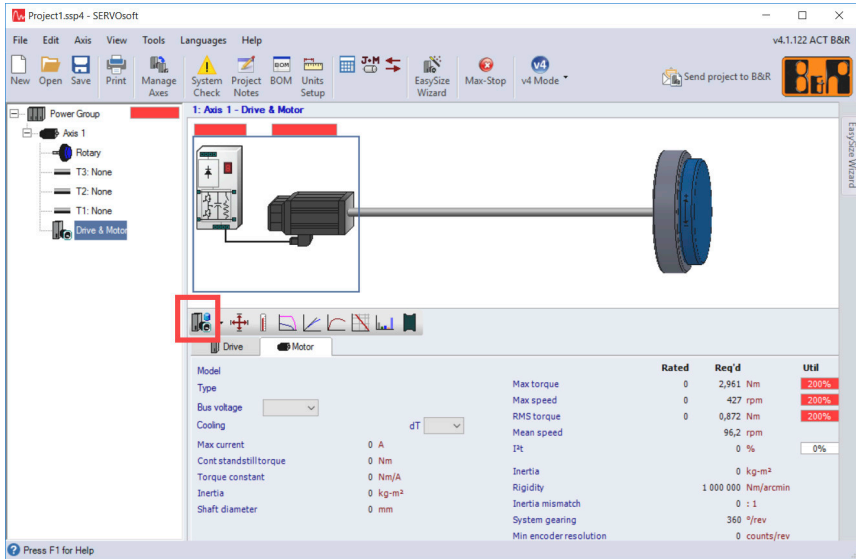
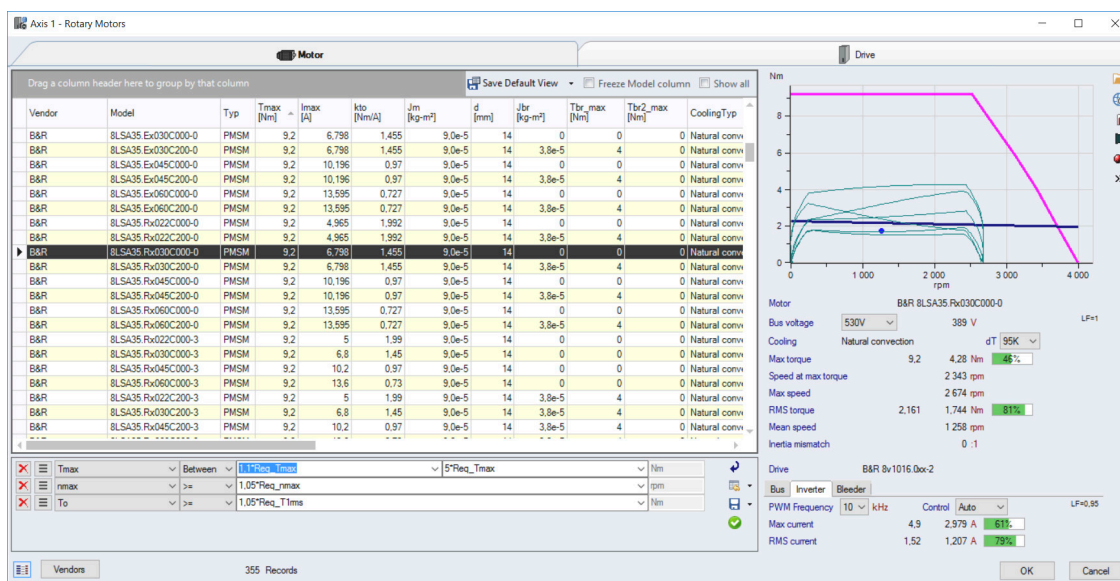
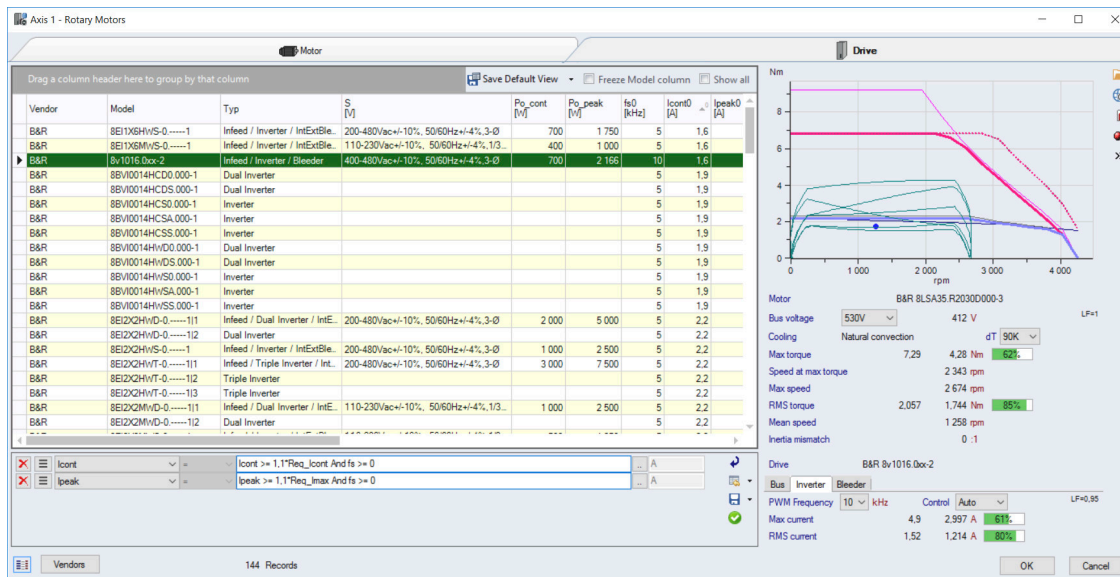


Figure 74: SERVOSOFT - Selecting drive units.

The motor and drive can now be selected. First, a motor suitable for the load profile is selected (see Fig. 75).



Afterwards, an inverter suitable for the motor is selected (see Fig. 76).



All percentage bars must be in the green area. When confirming the selection of inverter and motor, a summary is provided in the general overview (see [Fig. 77](#)). Also here, green percentage bars indicate that the drive unit can generate enough power for the required load profile or voltage and current for the motor.

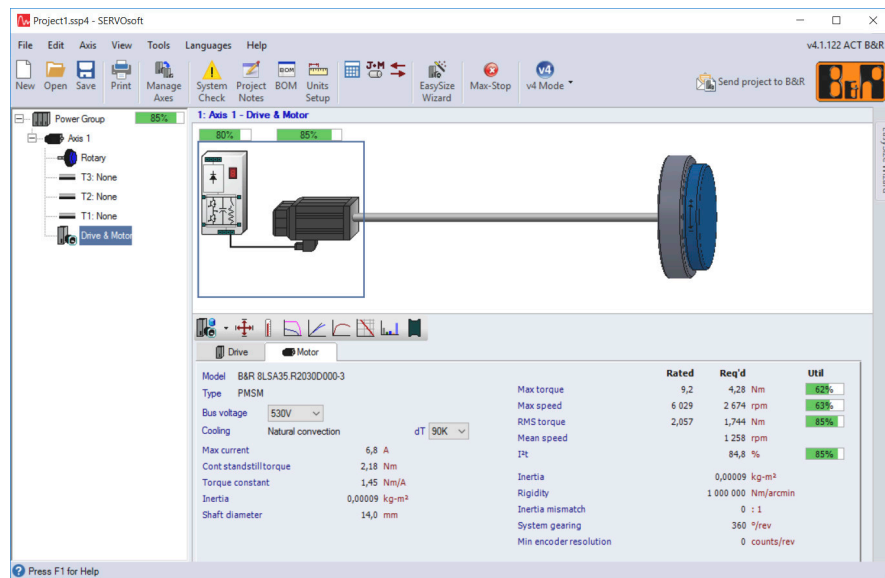


Figure 77: The SERVOnsoft software with correctly selected drive modules.

The selected modules can also be found on the B&R website, e.g. the ACOPOS module (see Fig. 78). This can now be integrated into B&R Automation Studio.

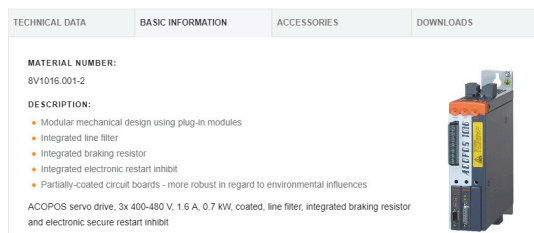


Figure 78: The selected ACOPOS module.

Additional information can be found in **TM465 - SERVOnsoft**.



TM465 - SERVOnsoft

Exercise: Using the SERVOnsoft Export toolbox

- Open the **SERVOnsoft** app.
- Load the system.
- Select the bridge drive.
- Set the output speed in rpm and select **thrust vs. velocity** in the output window. View the curve.
- Select a suitable directory and file name and export the data to the .xls file.

Exercise: Using the SERVOnsoft software

- Open the SERVOnsoft software.
- Select **Rotary** under the **Axis** group to load the load profile.
- Load the .xls file that was generated in MapleSim using the **B&R SERVOnsoft Data Generation** toolbox.
- Display and verify the corresponding speed and load torque curves.
- Now select the **Open servo motor database** option under **Drive and motor** to be able to size the drive unit.
- Select a suitable motor and inverter.
- Verify that the selected drive unit is the correct choice.

9 Adding an ACOPOS drive

Integrating an ACOPOS drive

The next step is to integrate an ACOPOS drive in B&R Automation Studio and simulate it parallel to the FMU for the overhead bridge crane. The speed of the ACOPOS drive is selected as the input parameter, which is then directly assigned to the speed of the overhead crane in the program code. To do this, an **Axis** object must first be created (see Fig. 79).

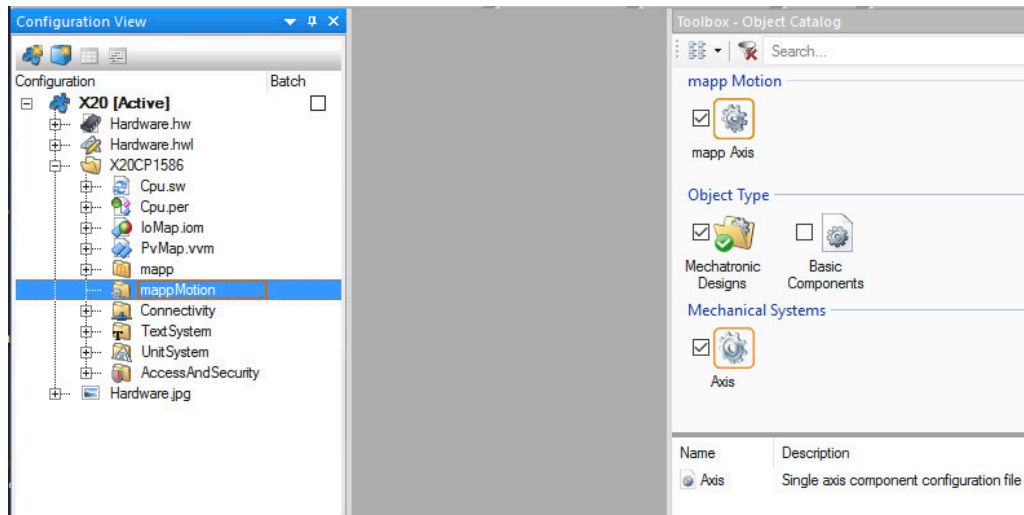


Figure 79: Adding an Axis object to the mappMotion library.

The **Axis** object must be configured accordingly (see Fig. 80).

Aktuelles 'axis' Element: gAxis_1			
Name	Wert	Einheit	Beschreibung
gAxis_1			
Base type	Rotary		Definiert die grundlegenden Bewegungseigenschaften der Achse
Measurement unit	Gradians		Maßeinheit für die Achse
Measurement resolution	0.01	Measurement	Mögliche Auflösung die für die Maßeinheit erreicht werden kann
Count direction	Standard		Bewegungsrichtung der Achse in welche der Positionswert ansteigt
Movement limits	Internal		Verschiedenen Grenzwerte die für Achsbewegungen berücksichtigt werden
Velocity	Basic		Grenzwerte für die Geschwindigkeit der Achse
Acceleration	280	Measurement	Geschwindigkeitsgrenzwert in jede Bewegungsrichtung
Deceleration	Basic		Grenzwerte für die Beschleunigung der Achse
Deceleration	50.0	Measurement	Beschleunigungsgrenzwert in jede Bewegungsrichtung
Deceleration	Basic		Grenzwerte für die Verzögerung der Achse
Deceleration	50.0	Measurement	Verzögerungsgrenzwert in jede Bewegungsrichtung
Alarms	None		

Figure 80: Configuring the Axis object.

The corresponding drive (found with SERVOSOFT) can now be added (see Fig. 81). The corresponding motor can also be integrated at the same time.

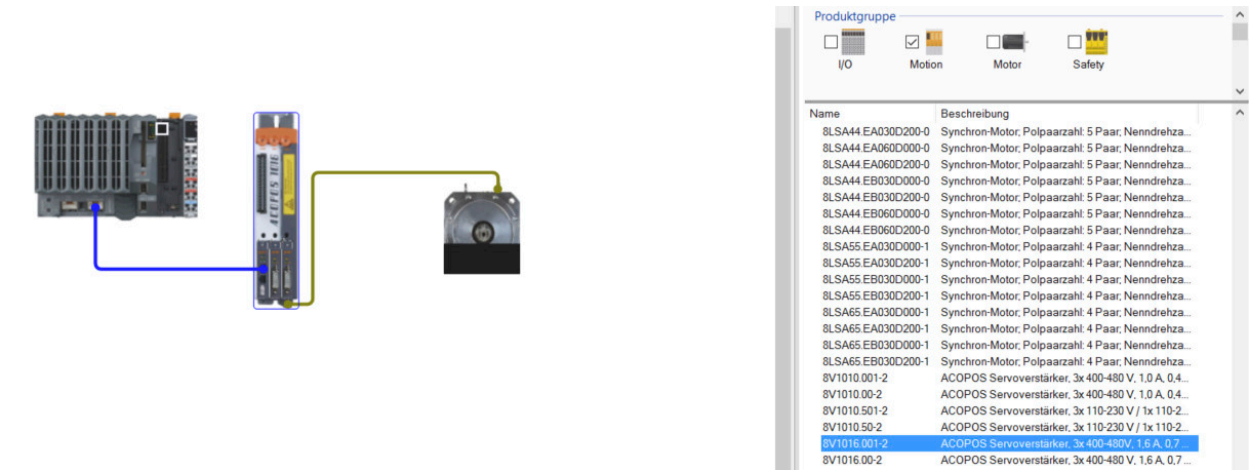


Figure 81: Adding an ACOPOS drive and a drive unit to the project.

The configuration of the ACOPOS drive is now opened and the defined **Axis** object assigned there (see Fig. 82, **gAxis_1**).

Name	Wert	Einheit	Beschreibung
8V1016.001-2			
Drive configuration			
Module			Contains parameters which influence or are relevant just for the whole module
Activate ACOPOS simulation ...	On		Activates or deactivates the ACOPOS simulation on the PLC
Power supply	AC		
Channel 1			
Real axis			
Axis reference	gAxis_1		(Name of the referenced axis component)
Mechanical elements			Parameter of hardware elements situated between motor encoder and load which influence the scaling
Encoder link	One encoder		
Motor and position en...	Encoder SS2...		
Controller			Axis controller parameters
Homing			Homing mode and parameters which can be used within the application program as preconfigured setting
Stop reaction			Reactions of the axis in case of certain stop conditions
Movement error limits			Limit values that result in a stop reaction when exceeded
Jerk filter	Not used		Jerk filter
Digital inputs			Various digital input functionalities e.g. like homing switch or triggers
Simulation			Parameters which influence the simulation possibilities of this axis
Axis features			Features für die Achse
Feature reference 1			Name der Achsfeaturereferenz
Virtual axis	Not used		
Channel features			Features für den Kanal eines Moduls
Feature reference 1			Name der Achsfeaturereferenz

Figure 82: Parameter values for the ACOPOS drive.

The configuration of the ACOPOS is thus completed. To be able to use this in conjunction with the FMU, a new program will be created (see Fig. 83).

Objektname	Beschreibung
Hallenkran	
Global.typ	Globale Datentypen
Global.var	Globale Variablen
Libraries	Globale Bibliotheken
operator	This library contains function interfaces for IEC 61131-3 oper
runtime	This library contains runtime functions for IEC tasks.
asTime	The As Time Library supports DATE_AND_TIME and TIME
AslecCon	This library contains function interfaces for IEC 61131-3 conv
sys_lib	The SYS_LIB library contains functions for memory manage
McBase	[mapp Motion] Base library
MpAxis	[mapp Motion] Axis library
McAxis	[mapp Motion] PLCopen axis library
McBase	Core library for mapp components
McAcpAx	[mapp Motion] ACOPOS axis library
CraneFmu	
CraneLib	
Types.typ	Exported data types
Constants.var	Exported constants
CraneLib.fun	Exported functions and function blocks
fmi2.cpp	
fmi2Functions.h	
fmi2FunctionType...	
fmi2TypesPlatfor...	
Crane	
Crane.c	
Crane.scn	
CraneFmiSce...	
CraneImpl.c	
fmuTemplate.h	
CADdata	
CranePrg	
Cyclic.c	Cyclic code
Init.c	Initialization code
Exit.c	Exit code
Types.typ	Local data types
Variables.var	Local variables
Program	
Main.st	Init, cyclic, exit code
Types.typ	Local data types
Variables.var	Local variables

Figure 83: Create a new task to connect the ACOPOS drive to the FMU.

In the new program, the speed output of the ACOPOS drive will be assigned to the speed input of the bridge (see [Fig. 84](#)).

```

PROGRAM _INIT

    MpAxisBasic_0.MpLink := ADR(gAxis_1);
    MpAxisBasic_0.Enable := TRUE;
    MpAxisBasic_0.Parameters := ADR(AxisParameters);

    AxisParameters.Position := 100.0;
    AxisParameters.Velocity := 10.0;
    AxisParameters.Acceleration := 50.0;
    AxisParameters.Deceleration := 50.0;

END_PROGRAM

PROGRAM _CYCLIC

    cranel.vel_BRIDGE := MpAxisBasic_0.Velocity;
    MpAxisBasic_0();

END_PROGRAM

PROGRAM _EXIT

    MpAxisBasic_0.Power := FALSE;
    MpAxisBasic_0.Home := FALSE;
    MpAxisBasic_0.MoveAbsolute := FALSE;
    MpAxisBasic_0.MoveAdditive := FALSE;
    MpAxisBasic_0.MoveVelocity := FALSE;
    MpAxisBasic_0.Stop := FALSE;
    MpAxisBasic_0.ErrorReset := FALSE;
    MpAxisBasic_0();

    MpAxisBasic_0.Enable := FALSE;
    MpAxisBasic_0();

END_PROGRAM

```

Figure 84: Program code to connect the ACOPOS drive output to the FMU input.

The drive can now be set in motion in Automation Studio to check the functionality. It is also possible to view the movement in the Scene Viewer file at the same time.

Additional information can be found in **TM412 - ACOPOSinverter configuration and commissioning**.



TM412 - ACOPOSinverter configuration and commissioning

Exercise: Integrating an ACOPOS drive into B&R Automation Studio

Creating an ACOPOS drive in B&R Automation Studio and testing it. A trace can be used to view the initial values and thus check the correct behavior of the overhead bridge crane.

- Create an axis in mappMotion and configure it.
- Insert an ACOPOS drive and a motor according to the configuration in SERVOfsoft.
- Configure the ACOPOS drive and link the axis that has been created to it.
- Create a new program and link the speed from the output of the ACOPOS drive to the input of the FMU.
- Give various commands to the drive and record the speed and motor torque. Observe the behavior of the model in the Scene Viewer.

11 Summary

Nowadays, it is possible to simulate extensive and complex machines and their physical properties.

It's therefore possible to simulate machines in advance without having to build the real machine. This saves time and money because errors can be detected and corrected at an early stage. The control functionality when using a B&R PLC for such machines can also be tested and verified in advance.

MapleSim is a program that allows simulation of physical entities at machine level. The Modelica approach allows even large and complex systems to be set up quickly and easily. This can be used to derive an FMU that can be integrated and simulated in B&R Automation Studio. The result can be visualized and verified using B&R Scene Viewer.

Drives can also be optimally sized to handle dynamic system conditions using SERVOSOFT. The **B&R SERVOSOFT toolbox** provides an easy way to quickly get the associated data into the program. An inverter and drive layout can be created. In B&R Automation Studio, the inverter and the drive unit can be simulated and coupled with the simulation model. This allows the layout to be verified.

The **MapleSim and Functional Mockup Interface (FMI)** training module provides initial insight into how physical systems or machines can be modeled as mathematical models using a physics modeler (MapleSim) and connected to models (drive model) in B&R Automation Studio.

MapleSim Gallery

MapleSim Gallery offers a large number of finished models. It is in any case useful to get ideas before modeling is started. This helps avoid aberrations. The models can be downloaded, and a video can also be used to quickly identify system behavior in advance.

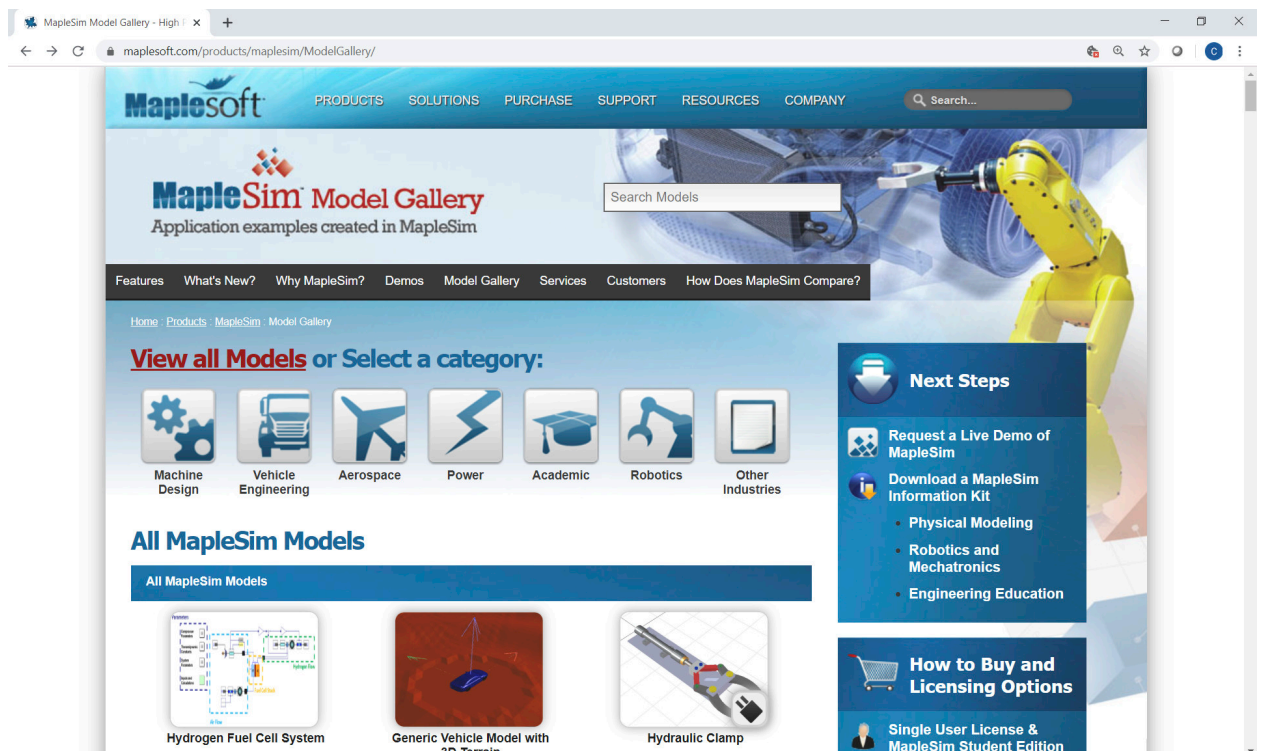


Figure 86: MapleSim Gallery

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B&R
Industrial Automation GmbH
A member of the ABB Group
B&R Straße 1
5142 Eggelsberg, Austria
office@br-automation.com

t +43 7748 6586-0
f +43 7748 6586-26

br-automation.com

